

MASTER

A Comparative Study of Explainable AI Methods for Computer-Aided Design

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A Comparative Study of Explainable AI Methods for Computer-Aided Design

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OMAPHTI-B: Master Thesis Project

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Abstract

Despite the growing body of research on AI-assisted decision-making and Explainable AI (XAI), generalizing findings to real-world applications remains a challenge. Studying underexplored decision tasks and developing domain-specific XAI methods can help diversify the field of XAI. Computer-Aided Design (CAD) tasks within the engineering domain present valuable research opportunities. This study addresses this by investigating the effect of XAI format on user experience and performance during CAD tasks. 32 participants engaged in simulated AI-assisted printed circuit board (PCB) design in an online experiment, with three conditions: no explanation, feature-based explanation inspired by SHAP values, and a novel explanation tailored to PCB design tasks named “Ideal Design”. Ideal Design combined visual and textual elements to express AI reasoning. Evaluation was conducted using behavioral measures, questionnaires, and qualitative feedback. Results showed significantly higher satisfaction for the feature-based explanation than for Ideal Design. The feature-based explanation better supported participants in the detection of bad AI suggestions. Qualitative results highlighted diverse individual preferences regarding which aspects of the explanation format were valued. Ultimately, this study provides insight into how explanations shape user experience for AI-assisted CAD tasks and the alignment of users, applications, and XAI. Ideal Design was suitable for providing a holistic understanding of AI reasoning, and the feature-based explanation supported critical evaluation of AI suggestions. These findings contribute to the advancement of XAI research by highlighting explanation trade-offs in a domain-specific context.

Keywords: AI-assisted decision-making, explainable AI, electrical engineering, computer-aided design, printed circuit boards

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Special thanks to my parents, whose care and support in every form, practical, emotional, and otherwise, enabled me to fully dedicate myself to this thesis. Their encouragement and belief in me meant more than words can say. Lastly, I would like to mention my family and friends for always showing interest in my work and offering support along the way. Their curiosity and encouragement have been a continuous source of motivation.

A Comparative Study of Explainable AI Methods for Engineering Tasks

AI-assisted decision-making is increasingly applied in the fields of healthcare, finance, and law (Lai et al., 2023). AI systems are believed to enhance the performance and efficiency of human-AI systems (Lai et al., 2023). AI-assisted decision-making can be described as AI providing predictions and/or explanations to assist in decision-making (Steyvers & Kumar, 2024).

Regulations such as the European Union's General Data Protection Regulation (Regulation (EU), 2016) and the Artificial Intelligence Act (2024) have established transparency and accountability as critical requirements for AI systems. This shows the recognition from governmental agencies that stakeholders must be able to understand how AI systems reach their conclusions. Similar sentiment can be found in the research field of human-AI Interaction (HAI) (Lai et al., 2023). The research field Explainable AI (XAI) is concerned with increasing transparency and trustworthiness of AI systems (Rawal et al., 2022). XAI is foundational for responsible AI, which emphasizes fairness, interpretability, and accountability in AI systems (Kim et al., 2024).

XAI research focuses on methods that provide explanations for AI decisions, including domains such as healthcare, finance, and law (Lai et al., 2023). Common XAI approaches include feature contribution and visual explanations, which are usually appreciated by end-users (Wang et al., 2022; Islam et al., 2022). Much research focuses on evaluating XAI methods to determine their suitability for particular tasks or application domains (Islam et al., 2022). Common evaluation metrics include trust, performance, mental models, and satisfaction (Kim et al., 2024; Lai et al., 2023). However, the field of HAI and XAI is relatively young and lacks standardization of its studies (Kim et al., 2024; Lai et al., 2023). It is thus important to select well-known, common, and high-quality evaluation methods.

Further findings from Lai et al. (2023) include that the researchers often study a limited set of decision tasks. Due to the limited variety of application domains in XAI research, challenges remain in developing XAI systems that effectively support across a broader set of domains and levels of user expertise (Wang et al., 2022).

One of those underexplored domains is that of engineering. While there is HAI research on AI within the engineering domain, such as that by Chong et al. (2022), no prior studies have evaluated XAI

methods specifically for engineering-related tasks. A potential application of (X)AI in the engineering domain is in the development phase. In this phase, engineers use Computer-Aided Design (CAD) software to design a technical product in preparation for manufacturing. CAD tasks involve open-ended problem solving, multi-step reasoning, and domain-specific constraints (i.e., electronics has different constraints than mechanics), resulting in tasks that differ significantly from classification or diagnosis tasks. In contrast to previously studied tasks (Lai et al., 2023), CAD tasks have unique characteristics: they follow the rules of physics and often requires calculations; have a theoretically ideal solution, but reality mandates making trade-offs; they take place in a 2D or 3D design space; have various levels of difficulty; have quality requirements that may or may not be documented; and come with professional risks. These characteristics present unique challenges and opportunities for XAI methods.

Engineers use complex mathematical algorithms in CAD tools to increase their efficiency and product performance. Mechanical CAD tools use computational fluid dynamics to optimize airflow through a computer (Lee et al., 1994); electronic CAD tools have algorithms called “auto-routers” that solve a maze problem for printed circuit boards (PCBs) to route interconnects between components (Badriyah et al., 2016); and architectural CAD tools perform structural analysis using finite element methods to compute building strength (Kaveh, 2014). A common thread among algorithms in CAD tools is that they are highly interactive, as they require human input and oversight to come to a satisfying design. Despite their valid output, their internal logic is often not transparent to users. This makes them an interesting evaluation task for XAI research, with potential generalizability across engineering domains. CAD tools for PCB design are specifically interesting for research due to their repeatable tasks that enable robust experimental designs. Additionally, the design problems have a symbolic structure (e.g., layers, traces, and net lists), which could facilitate rich explanations.

Unfortunately, the integration of algorithms in electronic CAD tools has faced severe challenges. Traditional auto-router algorithms are non-transparent in their decision-making, have not made significant technical advancements in the past decades, and the output is not satisfactory, so engineers still have to improve the output (Marrakchi, 2020). However, Zuken (2023) has recently shown promising results of their latest AI for PCB layout tasks. Instead of the traditional algorithms, it uses a black-box machine learning model, trained by reinforcement learning, and is based on verified high-quality

PCB designs. Even still, this commercial AI lacks explainability features, and the XAI field has not caught up to the developments of AI in CAD tasks.

This study contributes to the growing body of XAI research by investigating the underexplored engineering domain with a Computer-Aided Design evaluation task: AI-assisted PCB layout design. It explores how different explanation methods affect the user experience and performance of engineers. By doing so, this study addresses a crucial research gap and informs research and development of XAI systems to increase their suitability for engineering contexts. Therefore, this study proposes the following research question:

Which Explainable AI method gives engineers the best user experience in AI-assisted PCB design, and how does it relate to performance outcomes?

Literature Review

Explainable AI Overview

Explainable AI (XAI) refers to systems that make the internal reasoning of a black-box AI system more transparent and understandable to users. A central concept of XAI is the explanation, which may take on various forms depending on the AI model and task context. While various definitions of explanations exist, a widely cited and accepted definition is proposed by Miller (2019) based on social sciences. To Miller (2019), explanations are a cognitive and social process with the goal of helping users understand a decision. The cognitive process entails communicating the cause of an outcome, and the social process means the explanation should satisfy stakeholders. Thereby, an explanation should not only be accurate but also relevant, understandable, and pragmatic. This definition is suitable for HAI research as explanations should not only be technically accurate but also fit for complex real-world settings.

Various stakeholders are involved in AI development, each with different needs. Governments prioritize legal compliance, fairness, and accountability; developers require insights that aid in debugging, optimizing user experience, and aligning model behavior with intended outcomes; researchers seek valid, user-centered methods to evaluate explanation quality and address the lack of standardization in the field; end-users seek clarity, trust, and usability in their interactions with AI systems (Brdnik et al.,

2025; Hoffman et al., 2019; Nauta et al., 2023). Aligning XAI methods with the various stakeholder requirements can be challenging due to their diverse needs and discrepancies in XAI evaluation methods (Hoffman et al., 2019), but alignment is crucial to align with the EU’s vision for trustworthy AI (Regulation (EU), 2016).

XAI methods can often be grouped by their characteristics, as explained by Molnar (2025): post-hoc explanations take a black-box model and generate explanations after the prediction was made, whereas other AI models are interpretable by design (e.g., decision trees and linear regression models). Post-hoc explanations can be further divided into model-agnostic and model-specific explanations, where model-agnostic explanations can be applied to most types of AI models (e.g., partial dependence plots and counterfactual explanations) and the other is specific to certain types of AI models (e.g., saliency maps and adversarial examples). Model-agnostic explanations can be further classified into local explanations that clarify individual predictions or global explanations that elaborate on the overall model behavior. Other than these classifications, explanations can take on many forms (Molnar, 2025):

- Feature-based methods (e.g., SHAP and LIME), which quantify how much each input feature contributes to the model output.
- Example-based explanations (e.g., counterfactuals), which describe what inputs would give a certain output.
- Rule-based explanations (e.g., decision trees and decision rules), which explicitly describe the conditional logic that goes on in an AI model.
- Visual-based explanations (e.g., saliency maps), suitable for spatial and graphic data.

The large variety of XAI types creates the challenge of selecting a suitable XAI method given an AI model and application task. Poorly chosen or designed XAI methods can cause more harm than good. Harmful consequences have been identified to be overreliance, disengagement, dissatisfaction, or incomprehension (Steyvers & Kumar, 2024). The real-world application of such poorly designed XAI methods is risky to all stakeholders. It is therefore important to appropriately choose suitable XAI methods per application and evaluate whether these support alignment of AI, humans, and tasks.

Comparing the Benefits and Limitations of Explainable AI

Several studies have compared the effectiveness of different XAI methods across tasks and domains, thereby highlighting the benefits and limitations of each method. SHAP and LIME are two widely used and studied post-hoc methods. Salih et al. (2024) demonstrated that SHAP can explain non-linear feature relationships and has strong theoretical foundations in game theory. In contrast, LIME is more computationally efficient and model-agnostic but can only give local linear approximations, limiting insights into non-linear effects. Unfortunately, both methods produce inconsistent explanations depending on the used model and the presence of collinear features, which was found to be problematic in healthcare applications. While Salih et al. (2024) proposed a metric to quantify the explanation stability, they still found that minor differences led to major shifts in its feature's importance. Nwakanma et al. (2023) also compared SHAP and LIME for the application of presence detection. They found SHAP to be more intuitive thanks to its visual nature, while LIME was more accurate regarding its statistical predictions, and suggested including both to compensate for their weaknesses.

Rong et al. (2024) conducted a systematic analysis of 97 papers that conducted human-centered XAI evaluation. They found that XAI is typically effective in improving users' subjective understanding, but there are mixed results regarding improving trust and usability. Interestingly, interactive explanations tend to perform better in terms of user engagement and perceived usefulness. These interactive explanations can be questioned further or allow users to see changes in the explanation as they manipulate the input. Overall, it is clear that each XAI method has unique benefits and limitations. To match XAI with its application, they should be thoroughly evaluated.

Evaluation of Explainable AI

Creating coherent and goal-oriented XAI research requires careful consideration of how to evaluate different XAI methods. Kim et al. (2024) proposed a high-level distinction between human-centered and computer-centered evaluation. Human-centered evaluation focuses on assessing how explanations improve human performance and experience for decision-making tasks, whereas computer-centered evaluation of XAI emphasizes correctness and completeness of the explanations. The latter is

outside the scope of this study. Doshi-Velez & Kim (2017) further categorize human-centered evaluation into three approaches based on the level of how close the evaluation is to a real-world application:

- Functionally-grounded approach uses proxy tasks and does not use human users in its evaluation.
- Human-grounded approach uses simplified tasks and lay users to make a general evaluation.
- Application-grounded approach uses domain experts and real-world tasks to evaluate XAI methods for specific applications.

Each approach has strengths and weaknesses, and selecting an approach should depend on the research goal. For example, to evaluate the effectiveness of XAI in specific real-world domains (e.g., design or engineering), it is desirable to pursue an application-grounded approach.

Systematic reviews on XAI evaluation (Doshi-Velez & Kim, 2017; Kim et al., 2024; Lai et al., 2023) found that there is currently no consensus on which evaluation method should be used; the field of HAI is still developing and working towards standardization. Therefore, to promote comparison between studies, XAI research should use common evaluation metrics. Further recommendations from Kim et al. (2024) and Lai et al. (2023) include using multiple, complementary types of metrics to increase construct validity and encourage comparability across studies. It is thus beneficial to compare subjective and objective measures, as well as qualitative and quantitative.

Subjective measures such as self-reported questionnaires with Likert-scales and interviews and widely used to capture participants' perceived trust (Hoffman et al., 2023), satisfaction (Hoffman et al., 2023), and mental effort (Hart & Staveland, 1988). While these methods help capture the user's perspective, their limits are also acknowledged (van der Waa et al., 2021). Hoffman et al. (2019) note that subjective reports may not correspond with actual understanding and behavior. Thereby raising concerns regarding construct validity and misinterpretation of analysis results. A proposed solution to this limitation is to compare subjective and objective measures to see if, for example, self-reported ease of use aligns with objective performance measures (Lage et al., 2019).

Kim et al. (2024) and Lai et al. (2023) found common subjective measurements and methods to be based on the works by Hart & Staveland (1988), Hoffman et al. (2019), and Hoffman et al. (2023). A

survey of empirical methodologies by Vereschak et al. (2021) identified several trust-related behavioral measures, which this study will use to compare to subjective measures. Demographics such as years of relevant experience can be used as an exploratory predictor for performance-related evaluation.

Qualitative feedback on XAI interaction offers an opportunity to uncover rich and nuanced user needs. Qualitative evaluation methods include interviews, open-ended questions, focus groups, or the talk-aloud protocol. Though the most feasible method for online experiments is using open-ended questions, as they do not require a researcher to be present. Benefits from qualitative feedback include finding responses to specific XAI elements, working outside the bounds of single construct questionnaires, and gaining a deeper understanding of individual differences (Rong et al., 2024).

Computer-Aided Design as an Evaluation Task

AI-assisted engineering in CAD software comes with unique challenges and opportunities for HAI research. Especially when compared to the common set of evaluation tasks in HAI research such as prediction, classification, recommendation, and estimation tasks (Lai et al., 2023). CAD work requires exact solutions that are bound to physics. It involves technical problem-solving where engineers identify physical constraints to work around. It requires the use of domain knowledge, balancing trade-offs, and working in a 2D or 3D design space. Furthermore, while engineering decisions do not directly risk human lives, they carry significant consequences for product reliability, manufacturability, and safety. These characteristics are rarely found in other HAI evaluation tasks and provide the opportunity to evaluate technical reasoning and decision-making.

As was already mentioned, there is limited variety in HAI evaluation tasks. To the best of the researcher's knowledge, few studies have explored XAI for engineering tasks. One relevant study measured confidence in self and AI for bridge truss engineering tasks (Chong et al., 2022). In their experiment, participants connected bridge elements with trusses to make a strong construction. After participants made their decision, an AI suggested an alternative construction that the AI believed was optimal. After participants decided to follow the AI suggestion or use their initial design, they got feedback on whether their design choice was advantageous. Chong et al. (2022) found that the AI suggestions and

feedback could mislead users and inflate their confidence. Thereby highlighting the risks associated with using AI in critical tasks.

While no academic literature was found on XAI for engineering, some XAI methods are in use in commercial software. For instance, ESTECO's Reduced Order Modeling and ML toolbox generates physics-informed predictions of aerodynamic performance (Genovese & Scardigli, 2024). A "Proper Orthogonal Decomposition" is used to convert Computational Fluid Dynamics simulation data into interpretable components. These components are used in surrogate models, which make predictions for new designs. This method maintains physical traceability and explains how a change in design affects performance, making it suitable not just for experienced engineers but also those with less simulation expertise (Genovese & Scardigli, 2024). Finally, AI for creative design appears to be a growing field of research interest, such as applying generative AI to quickly make conceptual designs (Zhu & Luo, 2023). However, creative design should not be treated the same as engineering because the criteria for "correctness" are much more subjective for creative design.

Printed Circuit Board Design as an Engineering Task

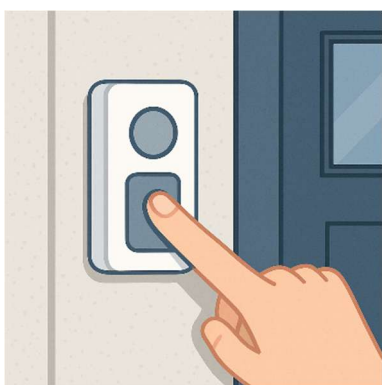
A sub-domain within engineering that is working towards AI integration is the development of printed circuit boards (PCBs). Similar to many other engineering domains, PCBs are developed on a computer in Computer-Aided Design (CAD) software. As explained in the introduction, CAD tools share many similarities across engineering domains, and studying a specific sub-domain could provide valuable XAI findings generalizable across engineering domains.

PCBs are a central component in electronic devices and can be found in practically all devices with a battery or power cord. A PCB consists of electronic components that can either be passive (e.g., resistors, capacitors, and inductors) or active (e.g., microcontrollers, sensors, and lights). Through the interplay of these components, a PCB gets a function tailored to the context it was designed for. For example, a PCB with a light sensor and a motor controller could function to close blinds when light is detected; alternatively, a PCB with a button, a microcontroller, a battery, and an antenna can work as a wireless doorbell system (see Figure 1.1). Figure 1.2 breaks down the wireless doorbell and shows an exploded view, as well as how interconnected electronic components work together to create a function

(see Figure 1.3). Figure 1.4 shows a render from PCB design software, and Figure 1.5 shows a view of the copper layer that PCB engineers typically work on. Finally, it is brought to the main topic of this study, an engineer creates the PCB in CAD software (see Figure 1.6).

Figure 1

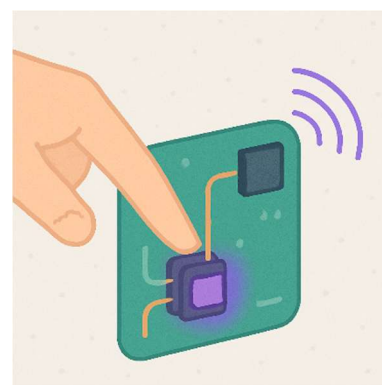
Doorbell System from User Interaction to PCB Engineering, subfigure 1, 2, 3, and 6 by OpenAI (2025)



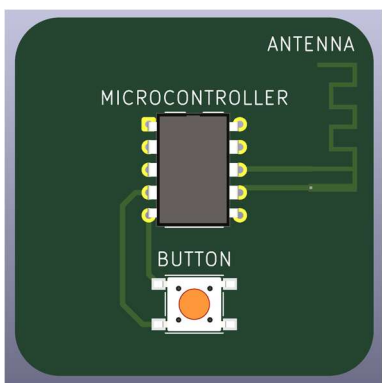
1. Pressing the button rings the doorbell. But how can it do that?



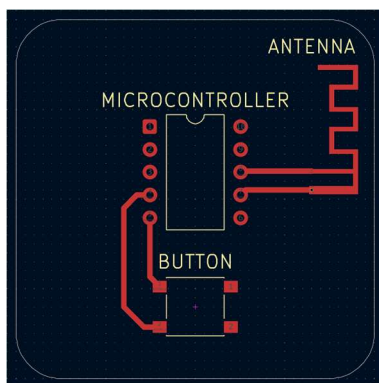
2. Inside is a circuit board with components that combine into a functionality.



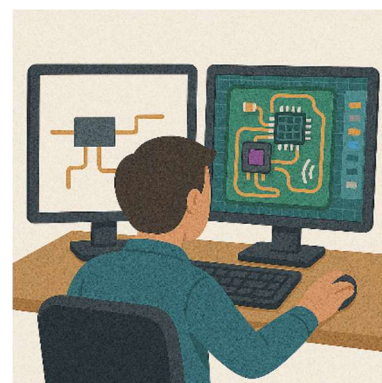
3. The button signal travels to the microcontroller, then is sent wirelessly by an antenna.



4. Copper traces on a circuit board interconnect component (screenshot 3D PCB render).



5. Designing requires consideration of trace routing and component placement (screenshot copper layer).



6. PCB engineers use ECAD software to design circuits before manufacturing.

The PCB engineering process is typically carried out by specialized engineers and requires several key steps, including component selection, schematic design, physical layout design, simulation, fabrication, and testing. Throughout the whole history of PCBs, engineers have had to manually place components and draw interconnections (“route traces”). Nowadays, CAD software is used, which automates many of the steps involved. Nevertheless, even in the most advanced CAD software, engineers

are still required to manually route traces using graphical interfaces. While auto-routers and auto-placers are available, their results are often unsatisfactory to engineers.

The company “Zuken” is one of history’s ECAD pioneers, and they recently developed an advanced AI model to do auto-routing and auto-placement (Zuken, 2023). This model is able to make high-quality designs with satisfactory results for engineers. AI implementation, such as that of Zuken, can significantly assist engineers in their design process, thereby saving much development time.

However, there is a lack of Explainable AI as the model's outputs are shown without explanations. This motivated this Master’s Thesis to further research how Explainable AI could be implemented into ECAD tools.

Current Research

Gaps between Current Explainable AI Methods and Engineering Requirements

Most XAI methods do not fully facilitate the requirements of PCB design. SHAP and LIME often rely on tabular, textual, or image data. However, PCB design software has a large variety of data types: text, geometrical, 2D, 3D, numerical, Boolean, and connectivity data. However, even these are not enough to capture all data in PCB design (Zuken, 2023). Given this large collection of data, popular XAI methods are unlikely to capture all the AI’s reasoning.

Additionally, XAI methods typically aim to explain a black-box prediction, which may not be as valuable to PCB engineers. Instead, engineers require nuanced support in their design and decision-making process. A domain-specific XAI method should go beyond showing why a prediction is made and instead clarify how a PCB design can be improved. It should understand the complex layout decisions, respect electrical and mechanical constraints, and allow engineers to remain in control. Without this domain-specificity, explanations risk being too abstract or generic to be meaningful.

A Conceptual Explainable AI Method for Printed Circuit Board Design

Given the aforementioned gaps and challenges in the field of HAI and the unique characteristics of PCB engineering, there is a need to develop a novel XAI method. This study introduces a concept for a novel XAI method tailored to PCB engineering and juxtaposes it with a well-known and studied alternative.

Fulfillment of several requirements is essential to be effective in the PCB domain. Given that PCBs include several sub-technologies, such as analog, digital, power, or serial signals, and component placement, the method must be flexible enough to adapt its content to the type of technology being designed. Due to the visual nature of PCBs, it should include at least visual, and preferably also textual, components to increase understanding in various user groups (Ribera & Lapedriza, 2019). Finally, due to time constraints, it was chosen to create global explanations to communicate the AI's internal rules and intentions globally. Even still, it should be “local-enough” to adapt to various technology types.

The novel XAI method's design was inspired by PCB “application notes”, commonly shared between manufacturers and electrical engineers to ensure designs are being created correctly. A global model-agnostic explanation is adapted to the PCB technology and offers justification for design choices made in the AI suggestions. The XAI method combines a visual diagram with textual statements to effectively inform users, and the content is based on books by Bogatin (2017). For example, when placing a component known as a decoupling capacitor, the XAI method will show an Ideal Design example, accompanied by texts like “place the capacitor as close as possible to the microcontroller”. Apart from mimicking application notes, junior engineers also receive similar feedback from senior engineers in the PCB domain.

To set up an effective XAI evaluation, the novel XAI method is juxtaposed with a contrasting method inspired by Shapley values. Shapley values are well-known and provide local model-agnostic explanations. In this study, the method is adapted into a Feature Importance waterfall plot. It shows how individual design features contribute positively or negatively to an AI's internal design score. Throughout the report, it is referred to as the “Feature Importance” XAI method, while the novel method is referred to as “Ideal Design”. While it is not made for engineering tasks, the Feature Importance is suitable enough to provide meaningful comparisons.

Aims and Contributions of the Current Research

This study aims to address the gaps in the field of HAI and XAI, especially around engineering tasks. Explainable AI is explored in the context of simulated but realistic PCB design decisions. The range of evaluation tasks in HAI is broadened by focusing on technical decision-making in the

engineering of PCB designs. An exploratory approach is chosen to compare a novel domain-specific XAI method with a well-established feature-based XAI method. A comprehensive evaluation is included by combining both objective measures (acceptance rate, error detection, decision time, and level of questioning) and subjective measures (trust, satisfaction, mental effort, and preference). Furthermore, qualitative feedback from participants is thematically analyzed to provide a rich insight into how users experience AI explanations. This study contributes to the development of context-specific XAI methods.

Method

Experimental Design

This thesis presents an exploratory study on decision tasks in AI-assisted PCB design and includes a novel XAI method, “Ideal Design”. A within-subject design was used with three conditions: first, a control condition (no XAI), followed by two counterbalanced XAI conditions, “Feature Importance” (similar to SHAP) and “Ideal Design”. The study aimed to follow the application-grounded approach (Doshi-Velez & Kim, 2017) but could not truly achieve this due to the challenges of integrating this study into a real work environment. Instead, an online study was conducted where participants were screened for relevant technical background, were given a simulated engineering task, and a performance-based scoring system with a monetary incentive to simulate a real-world engineering context. Following recommendations from the HAI field (Kim et al., 2024), standardized questionnaires were used to assess trust, satisfaction, and mental effort. To compare and gain a deeper understanding, additional metrics were collected: objective metrics (decision time, level of questioning, acceptance rate, and error detection) (Vereschak et al., 2021) and qualitative feedback on helpful and unhelpful XAI aspects. Finally, participants indicated their preferred XAI method. The experiment was hosted online on the TU/e HTI server and took no more than 50 minutes. The study was approved by the TU/e Ethical Review Board (experiment ID 2207), and all participants provided informed consent and received monetary compensation for their participation.

Participants

The sample size was calculated based on expected differences in the main dependent variable of interest: self-reported trust. A realistic and meaningful difference between any two conditions would

be 0.5 points on a 7-point Likert scale and a 1.53 points standard deviation (Brdnik et al., 2025). Using the Superpower package in R, a sample size of 25 is calculated to achieve a power of 90% (see Appendix A). Five extra participants were added to account for unforeseen issues. Therefore, the recruitment goal for this experiment was 30 participants.

Participants were recruited through the online platform Prolific (n.d.). After logging in, participants submitted the informed consent form (approved by the TU/e Ethical Review Board) and gave explicit consent to their participation and optionally consented to publicly sharing their recorded data in the Open Science Framework (n.d.). Prolific participants were recruited using digital screeners: a degree in engineering, age 18 to 99, Undergraduate, Graduate, or Postgraduate, approval rate of 95 to 100, and fluent in English. All recruited participants completed a pre-screening questionnaire with five questions and a limited time (see Appendix C). If participants had fewer than four answers correct or did not finish in time, they were blocked from continuing the experiment. Prolific participants were paid £6,50. Participants were informed they could win an extra reward (€3) if they were among the top 3 best-scoring participants. Participants who failed the pre-screening questionnaire still got a €0.70 compensation for their time and effort.

Seven TU/e electrical engineering students signed up for the experiment, though none of them responded to the online invitation. On Prolific, 76 participants responded to the experiment invitation, of which 19 failed the pre-screening questionnaire and 25 did not start, leaving 32 Prolific participants who completed the experiment. Only those who completed the experiment were included in the analysis. The total sample size was 32 ($N = 32$), with an average age of 32.9 years ($SD = 11.2$ years), ranging from 20 to 58 years old. The average years of relevant experience is 6.2 years ($SD = 4.1$ years), ranging from 1 to 20 years. Of the participants, 53.1% were males ($n = 17$) and 46.8% were females ($n = 15$). 56.3% had a Bachelor's degree (Undergraduate) ($n = 18$), 40.6% had a Master's degree (Graduate) ($n = 13$), and 3.1% answered "Other" in the education question ($n = 1$).

Stimuli

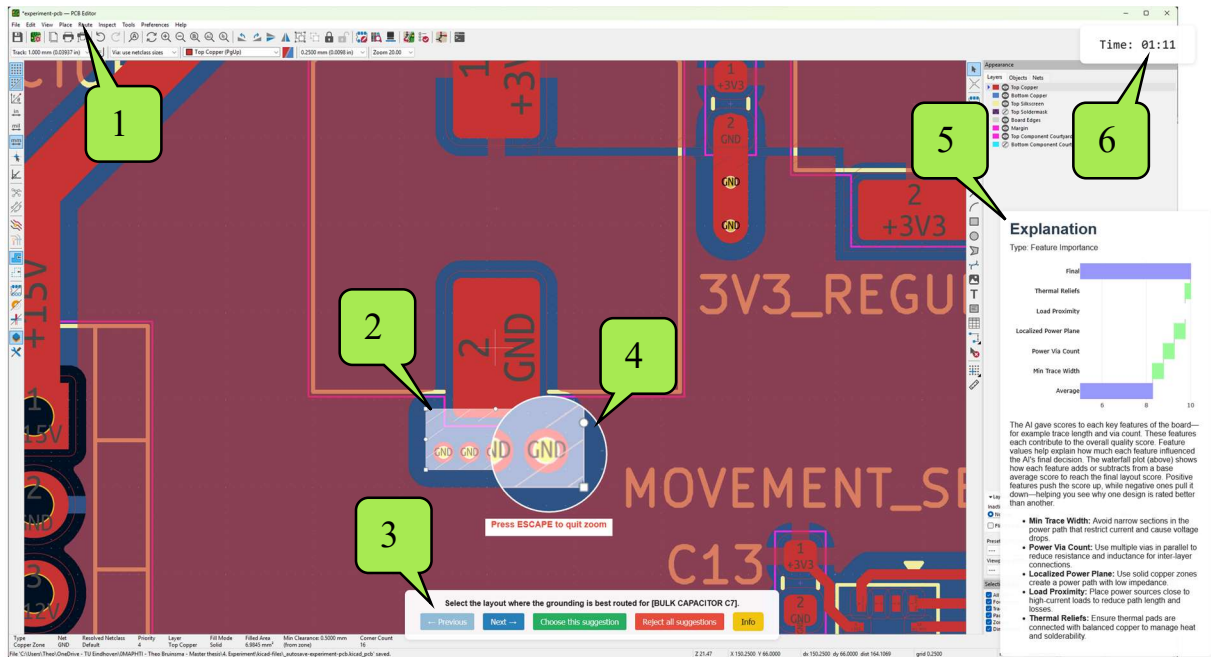
The stimuli used in this experiment were a series of interactive PCB design tasks presented on a custom web interface (see Appendix B). Each task came with a task description and four AI suggestions

(static screenshots of PCB designs). Participants could explore, accept, or reject suggestions by using the control buttons. A total of 39 unique tasks were created and randomly distributed across the three within-subject conditions: control (no XAI), Feature Importance (SHAP-like visual explanation), and Ideal Design (a novel XAI method modeled after technical application notes, providing both visual and textual explanations). Stimuli were consistent across conditions. For the 30 normal tasks, the suggestion order was ranked from good to suboptimal designs, whereas the nine error detection tasks featured four intentionally bad designs presented in no particular order.

Each condition came with a timer to simulate the time pressure found in a professional environment. Rejecting an AI suggestion triggered a time penalty of 3 minutes to simulate the cost of manually designing a PCB subsection. The task order was randomized to control for task difficulty and variance in suggestion quality. Participants' interactions were recorded per task, including decision time, key-strokes, and AI suggestion acceptance or rejection. The control condition was always presented first, while Feature Importance and Ideal Design were counterbalanced. Order effects may occur when participants always move from the control condition to an XAI condition. However, a different approach was deemed unfeasible due to practical constraints.

Web Interface

A new web interface was developed because no existing survey tool fit the experiment's purpose. The website can be run locally and requires PHP (version 8.4.7). Sent an inquiry to the researcher to request access (tpbruinsma@hotmail.nl). Figure 2 shows screenshots of the web interface. Participants could explore AI suggestions by clicking the buttons in the control window. With their engineering knowledge either accept an AI suggestion or reject all four suggestions. An overview of every window shown in the experiment can be found in Appendix B, including the login screen and questionnaires.

Figure 2*Screenshot of the Experiment Web Interface*

Note. 1) Screenshot of KiCad PCB design software; 2) Highlighted AI suggestion; 3) Control window; 4) Zoom functionality; 5) AI explanation window; 6) Timer.

Tutorial Video

After participants finished the pre-screening and gave their consent, they were presented with a tutorial video. The first tutorial introduced the experiment topic, the relevance of the study, and its general structure. A demonstration from Zuken (2023) illustrated the concept of AI-assisted PCB design in an enterprise software tool. Next, a technical background was given, participants were told they were designing a “robotic motor controller”, and were provided the relevant background knowledge (i.e., a block diagram, schematic, legend, and 3D renders). They were informed about the scoring system, which determined if they would receive a bonus reward. The goal of the study was not directly revealed. Instead, it was explained that the electronics industry is adding AI to their design software and that participating in the study helps with this development. Additionally, the EU’s regulations on the need for transparent and understandable AI systems were briefly noted.

After the control condition, participants viewed a second tutorial video explaining the addition of XAI elements. Starting with Ideal Design, which was described as similar to an industry-standard

“application note” (a technical document describing best design practices). It was described as explaining the goal towards which the AI optimizes (similar to a global model-agnostic explanation). Next, Feature Importance was explained as breaking down each AI suggestion into a waterfall plot. Herein, design features were valued as positive or negative, which resulted in a final design score, allowing for comparison among AI suggestions.

Engineering Tasks

Choosing one of the four AI suggestions is considered the engineering task in this study. Normally, during the PCB development process, engineers would have to manually place components onto a digital PCB (e.g., drag chips and resistors into position) and route traces between a component’s connection pads (e.g., draw connections where copper will remain after etching a copper clad). In these experiments, the development process was assisted by an AI that made decisions for placement and trace routing, thereby saving engineers a significant amount of time.

Engineering tasks can involve various aspects of the PCB, for example, routing a power supply line, a sensitive analog signal, or a noisy motor connection. Five variations of engineering tasks were chosen based on layout challenges that are commonly found in PCB designs (Salazar et al., 2019) (see Table 1). The engineering task type determined what circuitry and suggestions would be shown. Participants were tasked with selecting a design suggestion they deemed best, or could reject all suggestions.

Table 1*Overview of Engineering Task Types and the Corresponding Description Given to Participants*

Engineering task type	Task Description
Decoupling capacitor	Select the layout where the decoupling capacitor [C14] is best routed for chip [MOVEMENT_SENSOR1].
Power line	Select the layout where the power line is best routed to supply [INPUT 3.3V REGULATOR].
Digital signals	Select the layout where the digital signal [ROTATION_SENSOR OUTPUT A] is routed best.
Analog signals	Select the layout where the analog signal from [EXTERNAL TORQUE SENSOR] is best routed to [MICROCONTROLLER1].
Serial signals	Select the layout where the serial signals for [SPI MOVEMENT_SENSOR1] are routed best.

AI Suggestions

Each engineering task had four suggestions. A new database with AI design suggestions had to be created because currently, no such dataset exists. The experimenter created plausible and diverse solutions that an AI might suggest in a real-world setting. In the industry, multiple AI agents exist for this task, the AI designer made by Zuken (Zuken, 2023) being one of the best. Unfortunately, it was unfeasible to include in the experiment. Instead, the experimenter mimicked the AI interaction style seen in Zuken's demonstration videos and used their own PCB experience to create realistic designs. The suggestions were ordered based on design quality, from good to suboptimal. Except for the nine error detection tasks, where all suggestions were bad designs in no particular order.

It is possible that participants would show different behavior when presented with suggestions generated by an actual AI. However, the focus of this study is to take an interface without an XAI method and compare it with two XAI methods.

Error Detection Tasks

Per condition, three hidden errors were added to assess participants' reliance on AI explanations and accuracy of their mental model (Hoffman et al., 2019). In other experiments, these error detection tasks often focus on erroneous AI explanations (i.e., the explanation does not fit the suggestion). However, it was deemed more valuable to measure overreliance and whether XAI methods influence a participant's ability to notice bad AI suggestions. These tasks were added at random points after the third task for each condition. Participants were not informed about the presence of erroneous designs.

Scoring System

A scoring system was introduced to motivate participants to critically evaluate the design suggestions' quality and to simulate a professional PCB design environment. Professional PCB engineers pay close attention to make their designs "first time right", because mistakes can cost a lot of time and money. On the other hand, engineers have to manage their development time so as not to spend too much time. This can be generalized across engineering domains, as development comes with a tradeoff between design quality and development time. This tradeoff is mimicked by the scoring system, where it is better to accept good designs (AG), reject bad designs (RB), and spend less time (T). Total decision time per condition (excluding time penalties) was compared to the maximum observed decision time (T_{max}). The final formula is: $Score = \frac{AG}{30} + \frac{RB}{9} + \left(1 - \frac{T}{T_{max}}\right)$. Participants were informed during the tutorial video that they could win a bonus reward by following the aforementioned goals. The formula was not shared with participants.

Participants were given a three-minute time penalty for when they rejected an AI suggestion, which is roughly equal to the time spent manually designing a subsection of the PCB. This time penalty simulated a tradeoff that professional PCB engineers would have to make when using AI assistance in their design process (i.e., accepting AI suggestions saves time, but accepting bad designs lowers the score). The time penalties were not included in the scoring system, as the rate of wrongfully rejecting designs is already included in the formula in $\frac{AG}{30}$.

AI Explanations

The second and third conditions in this study were counterbalanced XAI methods: Feature Importance and Ideal Design. The latter is a novel XAI method developed specifically for PCB engineering tasks. Feature Importance explanations were adapted from SHAP and were included for their ability to map to PCB design features, give local model-agnostic explanations (i.e., motivate design choices per suggestion), and the extensive research available for comparison (Brdnik et al., 2025). The novel XAI method's Ideal Design gives a global explanation of what the AI optimizes for, not explained by variable names and values, but by given domain- and task-specific engineering rules. An AI assistant in PCB design can recognize the type of task at hand (i.e., Table 1), and the AI explanation will therefore change depending on the task at hand. This is also included in the experiment.

Feature Importance. This XAI method is adapted from the waterfall plot that is commonly used for Shapley values (Lundberg & Lee, 2017). A waterfall plot for Shapley values includes: an average score of the training data, weighted features, feature values, and a final predicted value. These elements can be mapped to the PCB engineering context, and for this experiment the following mapping was chosen: weighted features are measurable aspects of the PCB (e.g., component clearance or trace length), feature values are positive or negative depending on their contribution to the overall design quality, average score is the average design score from the AI's training data, and the final score is the summation of all values. However, there is no real AI in this experiment; all suggestions and explanations are made by the experimenter. The aforementioned values are therefore also made by the experimenter.

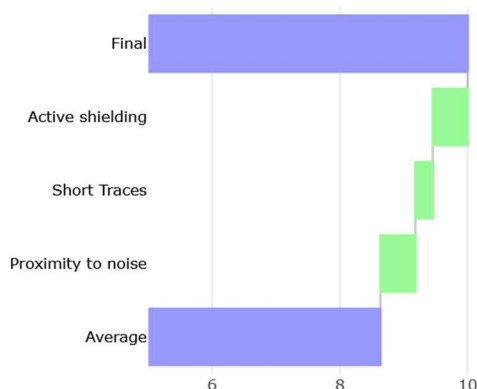
The features are chosen based on relevant design features for the given type of engineering task. For example, when routing an analog signal, features would include: proximity to noise, trace length, and active shielding. Figure 3 shows how the value of these features is plotted. This plot was shortly explained in the second tutorial video, and during the trials, a short textual description is given for the plot and its features.

Figure 3

Feature Importance Explanation for a Good Design (left) and a Relatively Worse Design (right)

Explanation

Type: Feature Importance

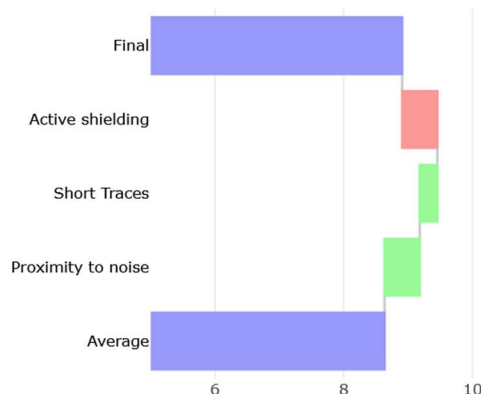


The AI gave scores to each key features of the board—for example trace length and via count. These features each contribute to the overall quality score. Feature values help explain how much each feature influenced the AI's final decision. The waterfall plot (above) shows how each feature adds or subtracts from a base average score to reach the final layout score. Positive features push the score up, while negative ones pull it down—helping you see why one design is rated better than another.

- **Proximity to noise:** Avoid routing analog signals near high-speed digital lines, clocks, or switching regulators.
- **Short Traces:** Short trace lengths reduce parasitic capacitance and inductance.
- **Active shielding:** Route a grounded or powered trace or planes alongside sensitive analog signals to block interference from nearby noise sources.

Explanation

Type: Feature Importance



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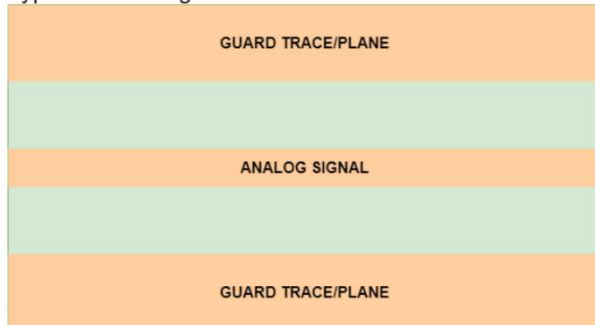
Ideal Design. This novel XAI method was developed specifically for PCB engineering tasks.

In contrast to Feature Importance, which uses plots, Ideal Design uses a visual explanation of what the AI optimizes towards. It shows a visual diagram and textual explanation containing detailed design guidelines: “Traces have a clearance of three times their width” (see Figure 4). Explanations are both textual and visual for general usability (Ribera & Lapedriza, 2019) and should increase correct understanding, team performance, ease of use, and appropriate trust (Szymanski et al., 2021). Ideal Design contrasts Feature Importance, which gives local explanations, whereas Ideal Design gives a global explanation.

Figure 4*Ideal Design Explanation for Analog Signals*

Explanation

Type: Ideal Design



- A. Traces have a clearance of 3 times their width.
- B. Only routed over continuous reference planes.
- C. Consistent trace width & minimized length throughout PCB.
- D. Guard traces/planes used for isolation.
- E. No vias used.

Feature Importance and Ideal Design have been chosen for their ability to be applied in the PCB context and their notable differences. These different explanations aim to see if differences between participants can be measured.

Measurements

Before the experiment's start, global information is collected: participant ID, login method, date, time, condition order, task order, and whether participants completed the pre-screening questionnaire.

Objective behavioral measures (Vereschak et al., 2021) were repeatedly measured for each task (both for normal and error detection tasks): decision time, level of questioning, final suggestion, and whether the AI suggestion was accepted. The decision time measurement starts when the engineering task is shown and stops when the participant submits their decision. The level of questioning measures how many suggestions the participant requests. The initial AI suggestion was always shown, and the participant could thus request three more AI suggestions. The final suggestion was recorded, and if this AI suggestion was accepted or rejected. Additionally, the condition in which the task was presented was recorded.

After completing all tasks of a given condition, questionnaires were presented to make a subjective evaluation of satisfaction (Hoffman et al., 2023), trust (Hoffman et al., 2023), and mental effort

(Hart & Staveland, 1988). For the control condition, only trust was recorded because the questionnaires for satisfaction and mental effort focused on AI explanations. Trust and satisfaction were measured by two multi-item 5-point Likert scale questionnaires, and mental effort with a single-item 20-point scale. The trust questionnaire included one reverse-coded item (Question 6), which was recoded before analysis. These evaluation methods are chosen as they are commonly used in the field of AI-assisted decision-making (Kim et al., 2024). After all conditions and questionnaires were completed, participants were asked to indicate their preferred XAI method and given four open-ended questions on what they found (un)helpful in each XAI method.

Finally, demographics were collected to gain a general understanding of the participants (age and gender), their relevant educational level, and years of electronics experience.

The internal consistency of the trust and satisfaction questionnaires was evaluated using Cronbach's alpha. All scales yielded alpha values above the commonly accepted threshold of $\alpha = 0.70$, indicating acceptable reliability (see Table 2). This suggests that the items within each questionnaire consistently measured the intended underlying constructs. For future analyses, item responses were averaged per participant and condition to create composite scores for trust and satisfaction.

Table 2

Cronbach's Alpha Tests for Internal Consistency per Condition

Measure	Condition	Cronbach's Alpha
Trust	Control	.789
	Feature Importance	.723
	Ideal Design	.812
Satisfaction	Feature Importance	.864
	Ideal Design	.899

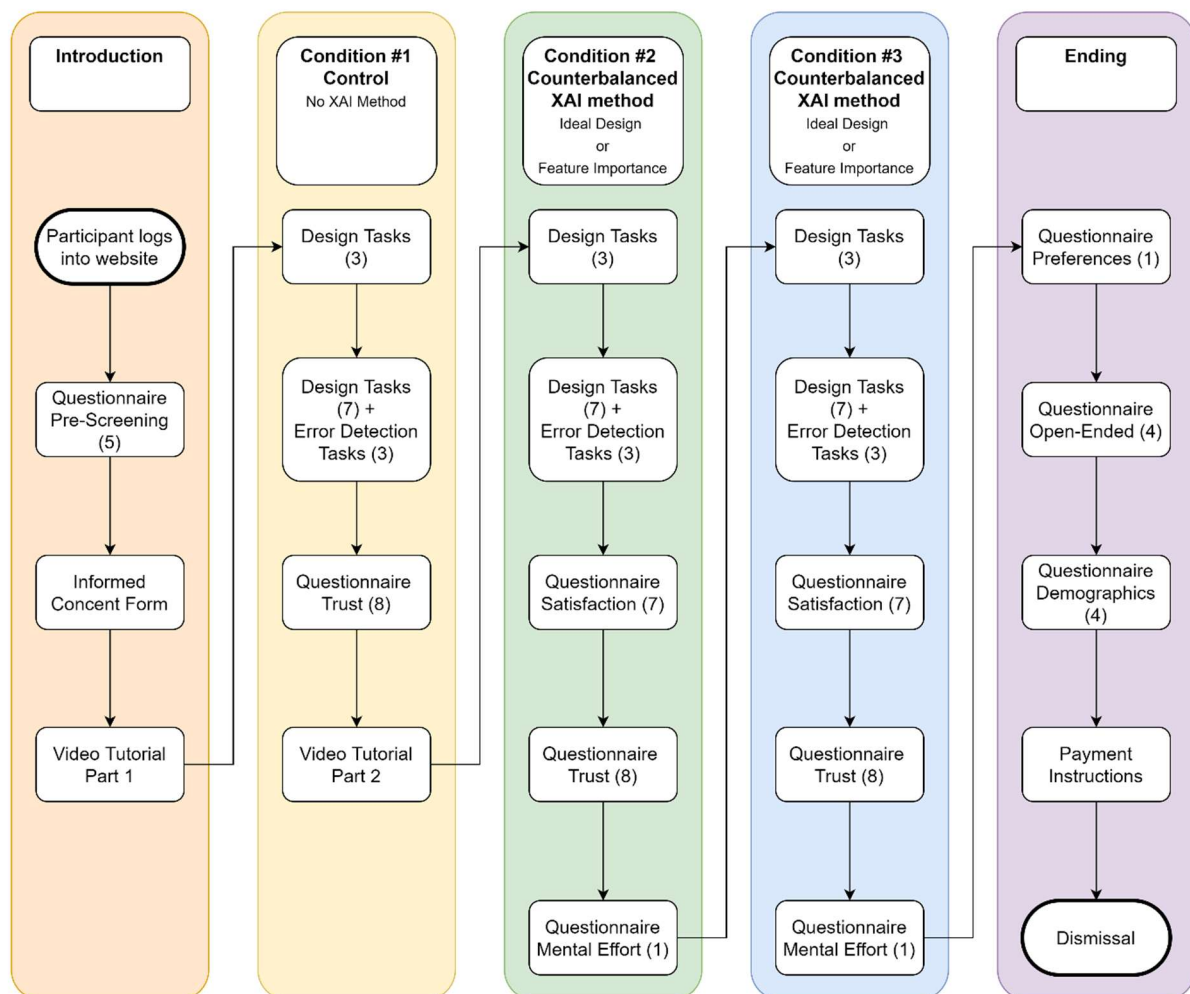
Procedure

The experiment was conducted online, and participants were advised to reserve 50 minutes and remain focused throughout the experiment. Figure 5 shows the full experimental procedure. A pre-

screening questionnaire was included to assess participants' relevant knowledge of PCB engineering (see Appendix C). This questionnaire included five multiple-choice questions and a limited completion time. The first nine participants had four minutes, which was then adjusted to three minutes for a stricter screening procedure. Participants who finished within the time limit and answered at least four questions correctly were allowed to proceed.

Figure 5

Experiment Procedure Flowchart



Next, participants were shown the informed consent form, which had been approved by the TU/e Ethics Review Board. After digitally signing the form, they were directed to the first video tutorial, which provided general information and instructions for the experiment. A link to the video can be found in Appendix C.

The experiment began with the control condition (i.e., no XAI method). Participants completed ten engineering tasks, wherein they used their engineering judgement to evaluate and select the best AI design suggestion (see Stimuli for more details). After the third engineering task, three error detection tasks were added at random positions throughout the remainder of the condition. These tasks included intentionally bad designs, and participants were expected to recognize and reject them. However, they were not explicitly told about these error detection tasks. Instead, the first tutorial stated that “the AI can suggest suboptimal designs” and “rejecting bad designs increases your score”. The first nine participants received no feedback after accepting or rejecting a design. But the remaining participants did receive feedback after accepting bad design suggestions to raise their awareness and critical thinking. After finishing 13 tasks in total, participants were given the trust questionnaire to assess their trust in the AI designer (see Appendix C).

The second video tutorial was then shown, which explained the addition of XAI methods with a short overview. Similar to the control condition, 13 tasks were given, but with the addition of an XAI method (Ideal Design or Feature Importance). After completing the tasks, three questionnaires were given (see Appendix C): satisfaction questionnaire about the explanation, a trust questionnaire about the AI designer, and a mental effort questionnaire about the task difficulty. Completing both conditions directed participants to the final webpage, where they indicated which XAI method they preferred. Four open questions about which aspects they (dis)liked and found (un)helpful allowed participants to motivate their decision. Demographics were collected, and then TU/e participants received payment instructions, whereas Prolific participants were directed back to Prolific, thus concluding the experiment.

Data Analysis

The data analysis was performed in Stata 17 (StataCorp LLC., 2025). Descriptive analyses were conducted for an initial impression of the measured variables.

Subjective Measures

A repeated-measures ANOVA was planned, with condition as a within-subject factor and participant ID as the subject identifier. This allowed for modeling individual variability in responses. Pre-model assumptions were evaluated. Normality was tested using the Shapiro–Wilk test as well as

skewness and kurtosis statistics. Equality of variance was examined with Levene's test. The assumption of normality was violated for trust, satisfaction, and mental effort. The cubic transformation improved the normality of trust and satisfaction sufficiently and allowed for a repeated-measures ANOVA on the transformed data. However, mental effort remained non-normally distributed, which was therefore analyzed using the nonparametric Kruskal–Wallis test on the non-transformed data. Equality of variance was tested with Levene's test and was not violated for any of the measures, both before and after transformation. Finally, preference data were analyzed descriptively and visually inspected across conditions but not included in other analyses.

Objective Measures

Acceptance and Rejection Rates. Acceptance and rejection rates were analyzed at trial-level for each participant. To conduct pre-model assumption tests, good and bad designs were analyzed separately, and the binary values at trial-level were transformed into decision rates at condition-level (only for assumption testing). Testing the rates revealed violations of normality and a negative skew for acceptance of good designs, while rejecting bad designs was positively skewed. Sphericity assumptions were not met either. Given the plan to analyze the binary outcome variables at trial-level, transformations were not attempted.

To analyze whether participants accepted good design suggestions and rejected bad design suggestions, a mixed-effects generalized linear model was used. This model could reveal trial-level variations while accounting for the highly skewed outcome variables by using a binomial distribution. Additionally, a logit link was selected to ensure predictions remain within a range of 0 to 1.

In both models, the XAI condition was included as a fixed effect (dummy-coded), and participant ID was modeled as a random intercept to account for individual differences. Additional covariates were explored for task- and participant-level variation: participant gender, age, years of experience, education level, condition order (binary variable “Ideal Design first”), and intervention exposure (if the participant got feedback on accepting bad designs). The trial number was not included due to high correlation with the XAI condition, the control condition was always presented first and always had the lowest trial numbers.

If the XAI condition was a significant predictor, post-hoc analysis of predictive margins would be conducted. The post-hoc assumption of outliers was assessed visually by inspecting histograms and box plots of the residuals. Observations showing extreme deviation were flagged as potential outliers. If there is a significant difference between Feature Importance and Ideal design, then

Decision Time and Level of Questioning. Pre-model assumption testing of trial-level measures revealed that decision time and level of questioning were positively skewed. Multiple transformations (e.g., log, square root, inverse) were applied, but did not improve normality. Kurtosis and sphericity assumptions were not met either. Because of the failed assumptions and complexity of trial-level variation, a mixed-effects generalized linear model, the same model as the decision rates.

For decision time, a gamma distribution with a log link was selected, as it is well-suited to skewed time-based measurements. For the level of questioning, which is an interval variable, a discrete Poisson distribution with a log link was used. In both models, the XAI condition was included as a fixed effect (dummy-coded), and participant ID was modeled as a random intercept to account for individual differences. Additional covariates were the same as for decision rates. Though an extra covariate was included to explore if bad design suggestions changed the outcome variables, a binary indicator for an error detection task was used. Additional interaction terms were included between error detection tasks and age, experience, and education level. Post-model analysis and assumptions were handled the same way as decision rates.

Summary of Qualitative Feedback

Participants' answers to the open-ended questions were summarized using a data-driven approach. Without prior expectations, responses were inductively coded based on their content. The following steps were taken: each response was given an initial code, similar codes were grouped, and each group was given a title and description that summarized the participants' feedback. Finally, the results were posted in a table with a frequency count of how often this group was mentioned. This analysis was manually conducted in Excel (Microsoft, 2025).

Results

Data Pre-Processing

Each participant completed ten standard engineering tasks and three error detection tasks per condition, resulting in 39 tasks in total. However, in one participant's datasets, seven trials were left unrecorded due to a software issue. Double-clicking the "Accept suggestion" button unintentionally skipped the next task. This issue was resolved immediately. Because this participant had consistent decision patterns throughout the experiment, it is assumed that their missing tasks behavior would have been comparable. Therefore, their data is still included in the analysis. A single participant with "Other" as education level was recoded to "Bachelor".

During a quality check of the first nine participants, it was found that most participants failed to reject any of the bad suggestions, thus failing the error detection tasks. This could indicate a lack of emphasis on the rejection option in the video tutorial and control interface. An intervention was introduced for the remaining participants: if a participant accepted a bad design, they received an alert stating, "Warning: This was a bad design! In future tasks, try to identify and reject designs with such issues.". Post-intervention data inspection showed that the rejection rate of both good and bad designs went up, potentially indicating increased awareness of bad design suggestions, negative consequences of accepting bad designs, and the choice to reject suggestions. The full dataset was used in subsequent analyses. Though results with and without pre-intervention participants were compared to assess the robustness of findings.

The aforementioned issue was not identified during the pilot study because the pilot participant correctly rejected seven out of nine bad suggestions. For the two bad designs that the pilot participant accepted, they reported that "I felt like I had to choose one of the designs, even if they were suboptimal. Only later I remembered that I could have rejected them". Before the experiment, it was thus not expected that error detection tasks would be missed.

Descriptive Statistics

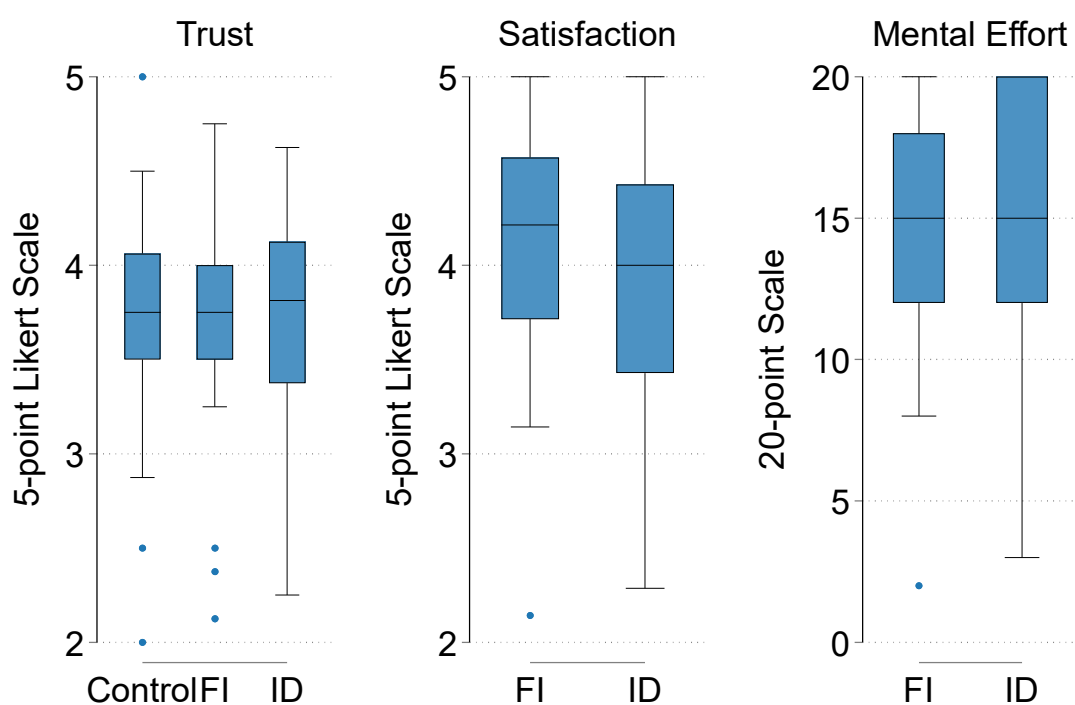
19 participants received the Feature Importance explanations before the Ideal Design explanation. The remaining 13 participants received the Ideal Design explanation before Feature Importance, a

true counterbalance was thus not reached. The cause was a software limitation: the condition order was alternated even when participants failed the pre-screening questionnaire. Resulting in an unequal balance for participants who did complete the experiment.

The trust scale, satisfaction scale, and mental effort results are shown in a box plot in Figure 6. Table 3 shows the descriptive statistics of all subjective measures (except preferred XAI method). Finally, 15 participants preferred Feature Importance, and 17 participants preferred Ideal Design.

Figure 6

Box Plot of Trust Scales, Satisfaction Scale, and Mental Effort per Condition



Note. Trust and satisfaction were measured using a 5-point Likert scale (5 = Strongly agree), where higher values indicate greater trust or satisfaction. Mental effort was measured on a 20-point scale, with higher values indicating greater perceived mental effort. FI = Feature Importance; ID = Ideal Design.

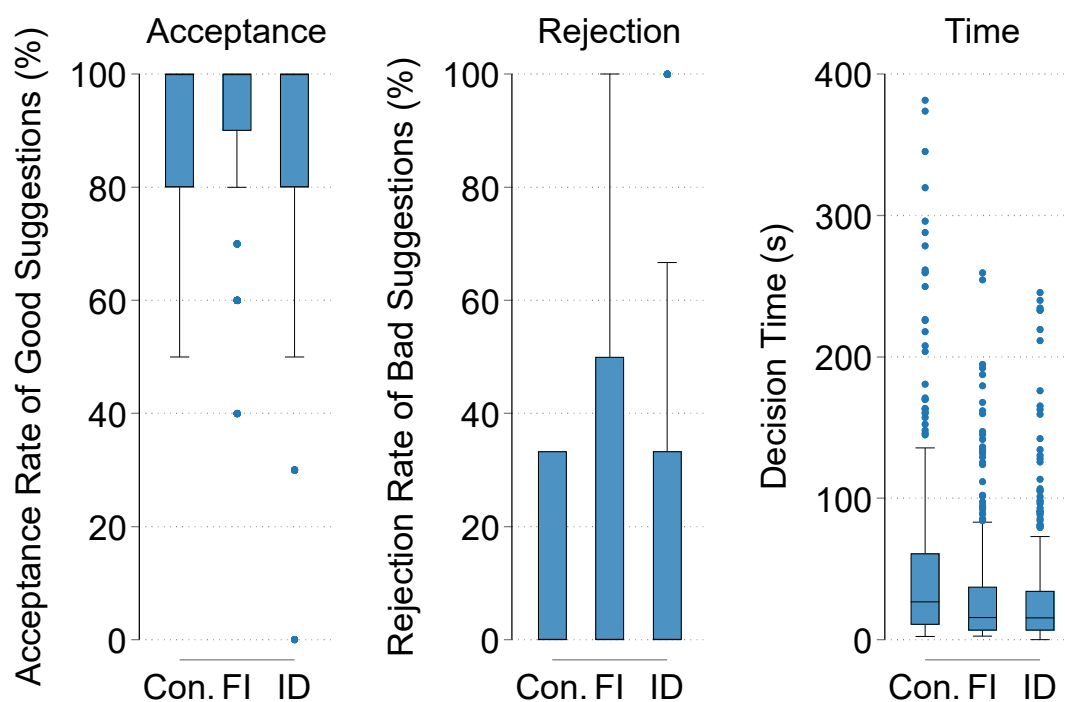
Table 3*Descriptive Statistics for Subjective Measures*

Measure	Condition	Obs.	Mean	SD	Min	Max
Trust	Control	32	3.719	0.591	2	5
	Feature Importance	32	3.699	0.570	2.125	4.75
	Ideal Design	32	3.684	0.655	2.25	4.625
Satisfaction	Feature Importance	32	4.152	0.609	2.143	5
	Ideal Design	32	3.911	0.746	2.286	5
Mental effort	Feature Importance	32	14.375	4.353	2	20
	Ideal Design	32	15.156	4.408	3	20

Box plots are used to visualize the acceptance rate of good designs, rejection rate of bad designs, and decision time (see Figure 7). For the level of questioning, a bar graph was used (see Figure 8). Table 4 shows the descriptive statistics of all objective measures. Acceptance and rejection rates were averaged across 13 tasks per condition, where a higher percentage indicates better decision-making. Decision time is a simple measure between when the task and suggestions are shown, and the decision is submitted. The level of questioning ranged between zero and three, zero indicating that participants did not request additional suggestions, and three indicating that participants requested all suggestions.

Figure 7

Box Plots of good design acceptance rate, bad design rejection rate, and decision time per condition



Note. Con. = Control; FI = Feature Importance; ID = Ideal Design.

Figure 8

Bar Plot of Level of Questioning per Condition

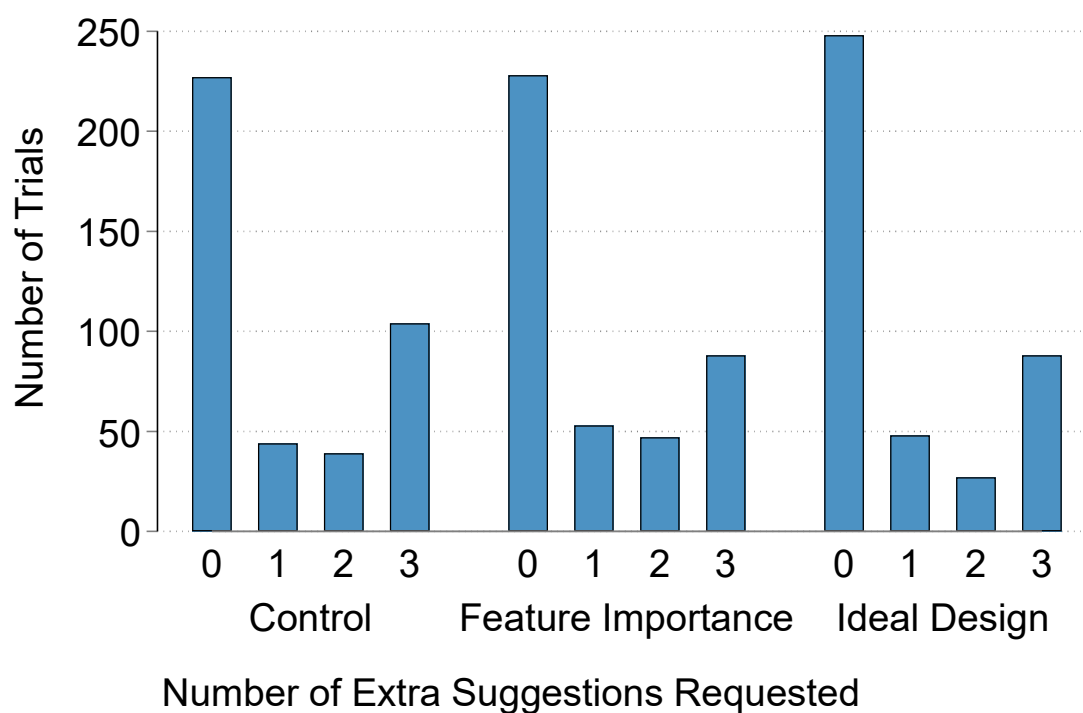


Table 4*Descriptive Statistics for Objective Measures*

Measure	Condition	Obs.	Mean	SD	Min	Max
Good Design Accepted	Control	319	0.887	0.317	0	1
	Feature Importance	320	0.903	0.296	0	1
	Ideal Design	317	0.886	0.318	0	1
Bad Design Rejected	Control	95	0.095	0.294	0	1
	Feature Importance	96	0.240	0.429	0	1
	Ideal Design	94	0.149	0.358	0	1
Decision Time	Control	414	58.017	109.165	2.2	1275.6
	Feature Importance	416	29.849	37.731	2.6	259.5
	Ideal Design	411	29.475	39.536	2.1	245.6
Level of Questioning	Control	397	1.065	1.287	0	3
	Feature Importance	403	1.010	1.240	0	3
	Ideal Design	403	0.901	1.240	0	3

The correlation between subjective and objective measures was explored without prior expectations. The nonparametric Spearman correlation table was used due to violations of normality in several measures. Correlations were explored with all conditions grouped together, but showed no significant correlations. Next, the correlations were tested for each condition separately (see Appendix D).

Two significant correlations showed up. In the control condition, the rejection rate of bad design suggestions correlated significantly with the trust scale ($\rho = -.378, p = .033$). Meaning that participants with a higher rejection rate of bad designs also showed lower self-reported trust during the control condition. In the Feature Importance condition, rejection rate of bad design suggestions correlated significantly with preferred XAI method ($\rho = .365, p = .040$). Meaning that participants with a higher rejection rate of bad designs also more often preferred the Feature Importance explanation.

Analysis of Trust Questionnaire

A repeated measures ANOVA was performed to compare the effects of three different XAI conditions and individual differences on self-reported trust (results in Table 5). The test showed that individual differences ($F(31, 62) = 10.37, p < .001$) was a significant predictor while XAI condition ($F(2, 62) = 0.08, p = .924$) was not. The model explained 84% of the variation in trust scores ($R^2 = .839$), with most variance contributed by between-subject differences. No post-hoc comparisons were conducted for the XAI condition, as there was no significant effect from the XAI condition. The analysis did not support any influence of XAI methods on trust.

Table 5

Subjective Trust Results of Repeated Measures ANOVA

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>p</i>
Participant	40961	31	1321.3	10.37	< .001
XAI Condition	20.173	2	10.087	0.08	.924
Error	7903.8	62	127.48		
Total	48885	95	514.59		

Analysis of Satisfaction Questionnaire

Descriptive statistics indicate that Feature Importance has higher satisfaction ratings ($M = 4.152$, $SD = 0.609$) than Ideal Design ($M = 3.911$, $SD = 0.746$). A repeated measures ANOVA was performed to measure the effect of XAI conditions and individual differences on self-reported satisfaction (see Table 6). The test showed that both individual differences ($F(31, 62) = 8.57, p = .009$) and XAI condition ($F(1, 62) = 7.55, p < .001$) are significant predictors. Individual differences explained far more variation than XAI methods. The model explained 90% of the variation in satisfaction scores ($R^2 = .898$), with most variance contributed by between-subject differences. No post-hoc tests were conducted as satisfaction was measured for two conditions.

Table 6

Subjective Satisfaction Results of Repeated Measures ANOVA

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>p</i>
Participant	54262	31	1750.4	8.57	< .001
XAI Condition	1541.0	1	1541.0	7.55	.009
Error	6328.4	31	204.14		
Total	62132	63	986.22		

Analysis of Mental Effort Questionnaire

Descriptive statistics reveal that Feature Importance has lower mental effort ratings ($M = 14.375$, $SD = 4.353$) than Ideal Design ($M = 15.156$, $SD = 4.408$). To check for significant differences, two Kruskal–Wallis tests were conducted. No significant effect was found for condition as predictor, $\chi^2(1) = 0.649$, $p = .421$ (corrected for ties). Participant ID as a predictor showed significant effects on mental effort, $\chi^2(31) = 56.295$, $p = .004$ (corrected for ties), indicating that self-reported mental effort differed significantly between participants. No post-hoc comparisons are conducted for the XAI condition as it only has two levels in this test.

Analysis of Accepted Good Design Suggestions

A mixed-effects generalized linear model with a binomial distribution and logit link was used to explore predictors of the binary variable “good design accepted” at the trial level ($N = 956$). The model included fixed effects for XAI condition, task presentation order, gender, intervention, condition order, age, years of experience, and education level. Error detection tasks were excluded from this analysis. A random intercept was specified for participant ID to account for repeated measures.

The main predictor of interest is the XAI condition, and the analysis was run multiple times: complete model, excluding demographics, excluding experiment variables (e.g., condition and task order), and finally, the model was run with only the XAI condition as a predictor. No significant differences were found thus the complete model is used.

Table 7*Mixed-Effects Generalized Linear Model for Correctly Accepting Good Design Suggestions*

Predictor	<i>B</i>	<i>SE</i>	<i>z</i>	<i>p</i>	95% <i>CI</i>
Feature Importance	0.207	0.288	0.72	0.472	[-0.357, 0.771]
Ideal Design	-0.001	0.280	0.00	0.997	[-0.550, 0.548]
Intervention	-2.638	0.835	-3.16	0.002	[-4.274, -1.002]
Ideal Design first	-0.336	0.627	-0.54	0.591	[-1.565, 0.892]
Female	-1.666	0.597	-2.79	0.005	[-2.837, -0.496]
Education: Master	0.815	0.685	1.19	0.234	[-0.527, 2.157]
Experience	-0.045	0.094	-0.48	0.634	[-0.230, 0.140]
Age	0.019	0.031	0.62	0.533	[-0.041, 0.079]
Participant ID	1.462	0.689			[0.580, 3.684]

The overall model was significant with Wald $\chi^2(8) = 18.70, p = .017$. Significant predictors were gender ($B = -1.666, SE = 0.597, z = -2.79, p = .005$) and whether the intervention was included ($B = -2.638, SE = 0.835, z = -3.16, p = .002$). From the model, it is suggested that females have accepted fewer good design suggestions, as well as the intervention having a negative effect. No post-hoc comparisons were conducted due to the main predictors not being significant.

Analysis of Rejected Bad Design Suggestions

A mixed-effects generalized linear model with a binomial distribution and logit link was used to explore predictors of the binary variable “bad design rejected” for error detection tasks on trial-level ($N = 285$). The initial model included fixed effects for XAI condition, gender, intervention, condition order, age, years of experience, and education level. Tasks with good design suggestions were excluded from this analysis. A random intercept was specified for participant ID to account for repeated measures.

The main predictor of interest is the XAI condition, and the analysis was run multiple times: initial complete model, excluding demographics, excluding experiment variables, and finally, the model was run with only the XAI condition as a predictor. Using only the XAI condition as a predictor resulted

in a significant model, and a significant post-hoc comparison. The final model therefore only includes the XAI condition as a predictor.

Table 8

Mixed-Effects Generalized Linear Model for Correctly Rejecting Bad Design Suggestions

Predictor	<i>B</i>	<i>SE</i>	<i>z</i>	<i>p</i>	95% <i>CI</i>
Feature Importance	1.380	0.479	2.88	.004	[0.441, 2.319]
Ideal Design	0.617	0.499	1.24	.217	[-0.362, 1.596]
Participant ID	2.411	1.313			[0.830, 7.008]

The overall model was significant with Wald $\chi^2(2) = 8.76, p = .013$. The only significant predictor was Feature Importance ($B = 1.380, SE = 0.479, z = 2.88, p = .004$). Its positive effect indicated that during the Feature Importance condition, participants were more likely to reject bad design suggestions. Post-hoc pairwise comparison analysis revealed that Feature Importance only had a significant contrast to the Control condition (Contrast = 0.139, $SE = 0.046, z = 2.86, p = .013$).

Table 9

Pairwise Comparisons of Predictive Margins for Decision Time

Pairwise Comparison	Contrast	<i>SE</i>	<i>z</i>	<i>p</i>	95% <i>CI</i>
Feature Importance vs Control	0.139	0.048	2.86	.013	[0.023, 0.254]
Ideal Design vs Control	0.051	0.042	1.23	.524	[-0.048, 0.151]
Ideal Design vs Feature Importance	-0.087	0.049	-1.78	.210	[-0.204, 0.030]

Analysis of Decision Time

A mixed-effects generalized linear model with a gamma distribution and log link was used to explore predictors of decision time at the trial level ($N = 1241$). The model included fixed effects for XAI condition, gender, age, years of experience, and education level. A random intercept was specified for participant ID to account for repeated measures.

The main predictor of interest is the XAI condition, and the analysis was run multiple times: complete model, excluding demographics, excluding experiment variables, and finally, the model was run with only the XAI condition as a predictor. No significant differences were found thus the complete model is used.

Table 10

Mixed-Effects Generalized Linear Model for Decision Time

Predictor	<i>B</i>	<i>SE</i>	<i>z</i>	<i>p</i>	95% <i>CI</i>
Feature Importance	-0.669	0.065	-10.29	< .001	[-0.796, -0.541]
Ideal Design	-0.705	0.065	-10.82	< .001	[-0.833, -0.578]
Intervention	0.218	0.256	0.85	.395	[-0.284, 0.719]
Ideal Design first	-0.043	0.247	-0.18	.861	[-0.527, 0.440]
Error detection task	0.582	0.194	2.99	.003	[0.201, 0.963]
Female	0.379	0.229	1.65	.099	[-0.071, 0.829]
Education: Master	-0.328	0.256	-1.28	.200	[-0.830, 0.174]
Experience	-0.015	0.039	-0.38	.702	[-0.091, 0.061]
Age	0.013	0.012	1.09	.274	[-0.010, 0.036]
Error * Education	-0.088	0.138	-0.64	.520	[-0.358, 0.181]
Error * Experience	-0.036	0.019	-1.9	.058	[-0.073, 0.001]
Error * Age	-0.012	0.006	-1.91	.056	[-0.025, 0.000]
Participant ID	0.362	0.096			[0.215, 0.607]

Twenty-nine outliers were found with *z*-scores above three, equivalent to ± 180 seconds. Their inclusion did not significantly affect model coefficients and *p*-values. Inspecting the outliers' keystrokes showed that some had extensive interactions, while others made no keystrokes, potentially due to being distracted. Moreover, 27 out of 29 outliers belonged to the Control condition. Therefore, excluding outliers should be interpreted with caution, as it may suppress variability inherent to the baseline condition. Given the risk of excluding realistic observations, all observations were included in the main analysis.

The overall model was significant with Wald $\chi^2(12) = 178.72, p < .001$. A significant and positive effect came from when the task was an error detection task ($B = 0.582, SE = 0.194, z = 2.99, p = .003$), showing how participants took longer to make a decision when the AI gave bad design suggestions. The remaining significant effects were negative and came from the XAI conditions. Feature Importance significantly decreased the decision time ($B = -0.669, SE = 0.065, z = -10.29, p < .001$), as well as Ideal Design ($B = -0.705, SE = 0.065, z = -10.82, p < .001$). Other factors and interactions were not significant.

To further understand the effect of each condition, a post-hoc pairwise comparison analysis was conducted with Sidak-adjusted significance levels. (Table 11). Both explanations had a significant contrast in comparison to the control condition. But compared to each other, the explanations were not significantly different. A possible interpretation is that both were effective at decreasing decision time, but this could also be due to learning or fatigue effects. For that reason, a linear fit with a 95% confidence interval was plotted with trial number on the x-axis and decision time on the y-axis (see Appendix E). This included both condition orders and revealed a clear downward trend in decision time, even when excluding the control condition tasks. Two margin plots were made, one for each condition order (see Appendix E). This similarly revealed a downward trend in decision time. It is therefore very likely that the decrease in decision time is not due to the explanations but due to learning or fatigue effects.

Table 11

Pairwise Comparisons of Predictive Margins for Decision Time

Pairwise Comparison	Contrast	SE	z	p	95% CI
Feature Importance vs Control	-28.763	4.621	-6.22	< .001	[-39.796, -17.730]
Ideal Design vs Control	-29.855	4.710	-6.34	< .001	[-41.102, -18.608]
Ideal Design vs Feature Importance	-1.092	1.893	-0.58	0.917	[-5.613, 3.428]

Analysis of Level of Questioning

A mixed-effects generalized linear model with a Poisson distribution and log link was used to explore predictors of decision time at the trial level ($N = 1241$). The model included fixed effects for

XAI condition, intervention, condition order, task type (error detection), gender, age, years of experience, education level, and interactions between error detection tasks and participant characteristics. A random intercept was specified for participant ID to account for repeated measures.

The main predictor of interest is the XAI condition, and the analysis was run multiple times: complete model, excluding demographics, excluding experiment variables, and finally, the model was run with only the XAI condition as a predictor. No significant differences were found when excluding variables. Thus, the entire model is reported.

Table 12

Mixed-Effects Generalized Linear Model for Level of Questioning

Predictor	<i>B</i>	<i>SE</i>	<i>z</i>	<i>p</i>	95% <i>CI</i>
Feature Importance	-0.056	0.069	-0.81	.416	[-0.191, 0.079]
Ideal Design	-0.168	0.071	-2.37	.018	[-0.307, -0.029]
Intervention	-0.171	0.667	-0.26	.798	[-1.479, 1.137]
Ideal Design first	-0.518	0.650	-0.8	.425	[-1.792, 0.755]
Error Detection Task	0.188	0.242	0.77	.438	[-0.287, 0.662]
Female	-1.913	0.612	-3.13	.002	[-3.112, -0.713]
Education: Master	-0.455	0.670	-0.68	.497	[-1.769, 0.858]
Experience	-0.041	0.103	-0.4	.689	[-0.243, 0.161]
Age	0.011	0.031	0.37	.714	[-0.049, 0.072]
Error Dete... * Education	0.041	0.157	0.26	.795	[-0.267, 0.348]
Error Det... * Experience	-0.028	0.028	-1	.320	[-0.083, 0.027]
Error Detection t... * Age	-0.001	0.007	-0.17	.866	[-0.014, 0.012]
Participant ID	2.496	0.792			[1.34, 4.648]

The overall model was almost significant with Wald $\chi^2(12) = 19.01, p = .088$. Ideal Design had a significant negative effect ($B = -0.168, SE = 0.071, z = -2.37, p = .018$), and female participants had

slightly higher levels of questioning ($B = -1.913$, $SE = 0.612$, $z = -3.13$, $p = .002$). Post-hoc margin analysis revealed no significant contrasts. Due to the lack of significance, this was not analyzed further.

Summary of Qualitative Feedback

Participants' answers of the open-ended questions were read and summarized, as described in the Methods section. A complete summary of the feedback can be found in Appendix . The paragraphs hereunder describe the two most common helpful and unhelpful aspects for each condition. All reported aspects will be considered for the Discussion section.

For Feature Importance, six distinct aspects were identified. The most often reported helpful aspect was that it explicitly connected design features to the AI's reasoning, thereby reportedly helping participants understand what PCB design aspects to focus on or improve (mentioned nine times). The second most common helpful aspect was that the explanation was easy to understand, and supposedly supported trust and mental processing (mentioned seven times).

Five distinct unhelpful aspects were reported for Feature Importance. The most common reason was that the explanation was too abstract or difficult to understand, participants reported that it was missing actionable guidance or examples (mentioned nine times). The second most common aspect is that the explanation was missing visual support and was reportedly harder to contextualize to the CAD environment (mentioned five times).

For Ideal Design, five distinct helpful aspects were identified. The two most frequently mentioned were that the explanation was clear and easy to understand, it reportedly helped simplify the complex design context (mentioned eight times), and secondly, that it provided concrete examples of good PCB layouts. This supported interpretation and application due to the visual nature (mentioned eight times).

Five unhelpful aspects were identified for Ideal Design. The most frequently mentioned issue was that the explanation lacked sufficient detail or clarity, participants reported that the content or visuals were unclear (mentioned five times). Another common concern was that it did not explain why certain layouts were preferred or what the consequences would be for suboptimal choices (mentioned four times). Similarly, some found the feedback too vague or generic to be actionable (mentioned four times).

Discussion

This study aimed to investigate the influences of explainable AI methods on user experience and performance during simulated AI-assisted PCB layout tasks in CAD software. For this aim, an online experiment was conducted wherein participants used AI design suggestions to create a printed circuit board. The combined measures suggest that the Feature Importance explanation supports participants' critical evaluation of design suggestions better than Ideal Design. Qualitative analysis revealed that Ideal Design was better for building a holistic understanding of the AI's reasoning, while Feature Importance was more helpful at comparing AI suggestions. Finally, variations in individual preferences highlighted different cognitive processes in participants and demonstrated how explanations should not only align with the application but also with individual users.

Influences of XAI Methods on Subjective Trust, Satisfaction, and Mental Effort

Comparison results of the trust questionnaires showed no significant differences between XAI conditions. This suggests that including an XAI method does not influence trust in the AI designer. Given that trials and their corresponding AI suggestions were randomized across conditions suggests it is unlikely that suggestion quality could have contributed to measurable differences. Instead, individual differences were likely the primary cause of variability in trust. Prior research has identified several factors that influence trust in AI systems, including users' prior experience with AI and individual self-confidence (Vereschak et al., 2021), which were not measured in this study but may explain the observed variation.

Comparison results of the satisfaction questionnaires showed significant differences, wherein Feature Importance had higher satisfaction than Ideal Design. Though this difference was relatively small compared to individual differences. This suggests that Feature Importance has a slightly better user experience than Ideal Design. Prior research has emphasized that user satisfaction with AI explanations can be determined by factors such as convenience, logical inference, and credibility, as well as perceptions of usefulness, ease of use, and the system's ability to adapt and improve over time (Xing & Jiang, 2024).

Comparison results of the mental effort scale showed no significant difference between the two XAI methods. Most variation was on participant-level, which could indicate that influences from familiarity with circuit board design, technical confidence, or task engagement have an influence on cognitive workload. Additionally, it is possible that the participants' answers to mental effort were directed to the task, and not to how difficult it was to understand the AI explanation.

Influences of XAI Methods on Decision Rates, Decision Time, and Level of Questioning

The model predicting acceptance of good designs was statistically significant, though the XAI condition itself was not a significant predictor in either case. Female participants showed significantly lower acceptance rates of AI suggestions, indicating a gender-related difference in AI reliance. This aligns with findings from Møgelvang et al. (2024) and Russo et al. (2025), who both found that females show less reliance on AI. The second significant predictor was the inclusion of the feedback intervention (i.e., an alert when accepting bad design suggestions), which significantly reduced acceptance rates. This could indicate that participants became more critical of AI suggestions when they received an alert that they accepted a design they should have rejected.

The rejection of bad designs showed a significant model. Feature Importance was a significant predictor and showed that participants were better able to detect bad design suggestions with this explanation. This may be due to the local explanations that quantified bad design aspects and the feature contribution bars becoming red for badly executed design features. Whereas the global Ideal Design did not change when exploring suggestions for a design task. Similar findings were found by Prinster et al. (2024) in healthcare, where local explanations improved diagnostic accuracy more than global ones.

A significant model was made for decision time. Analysis showed a significant positive effect for bad design suggestions, indicating that participants recognized these bad designs and were more critical during their decision-making. However, descriptive statistics show that the rejection rate of bad design suggestions was typically below 50%. This means that while the decision time was typically higher, it did not always lead to successfully rejecting the bad design suggestions. A significant negative effect came from the Feature Importance and Ideal Design XAI conditions. Post-hoc comparison showed significant differences between both conditions and control, but not between each other. Visual

inspection for learning or fatigue effects showed that decision time decreased throughout the experiment (Appendix E). The reduction in decision time should be interpreted carefully and may largely be due to learning or fatigue effects.

The model for the level of questioning was almost significant and came with a non-significant post-hoc comparison. It is therefore unlikely to contain useful information.

Exploratory correlation analysis showed two significant correlations between subjective and objective measures. The rejection rate of bad designs negatively correlated with self-reported trust (for the control condition). This may indicate that participants adapted their strategies based on the available XAI method. For example, they may have relied more on their judgment in the control condition.

The preference for Feature Importance positively correlated with the rejection rate of bad designs. This may indicate that users who favored local feature-based explanations were better able to apply them to effectively reject bad designs. This aligns with the results of the mixed-effects generalized linear model used for rejections of bad design suggestions, where Feature Importance had significantly higher rejection rates than the control condition. These findings may demonstrate the ability of local explanations to better demonstrate that certain design aspects are executed poorly by the AI.

Comparing Qualitative Feedback Between XAI Methods

Feature Importance was appreciated for clearly highlighting which PCB properties influenced the AI's decisions and in which direction, thereby supporting design prioritization and improvement. However, several participants noted that it felt abstract and was sometimes difficult to translate into concrete design actions. Salih et al. (2024) also found that feature-based explanations are typically more difficult to grasp. In contrast, Ideal Design was frequently described as easy to understand, visual, and intuitive. This was attributed to the visual explanation being congruent with the appearance of a real PCB layout. Participants found it helpful to see what the AI considered a good design, which supported faster comprehension and action. That said, some feedback pointed to Ideal Design being overly general and lacking in detailed justifications. For example, it did not explicitly explain why the first suggestion is preferred over the second suggestion, and what performance consequences each has. In sum, Feature Importance tended to support understanding and prioritization of PCB properties, while Ideal Design

facilitated holistic understanding and practical application, though it lacks some precision and depth. These patterns suggest that the two methods serve different cognitive needs and user experience may depend on the user's goals.

Implications

This study has several theoretical, practical, and methodological implications for research and development of explainable AI in the engineering context and AI-assisted PCB design. Theoretically, the results support existing literature that emphasizes the significance of individual user characteristics for shaping trust in AI systems (Vereschak et al., 2021). While previous research often focused on generalized tasks such as classification or recommendation, this study contributes insights into how individual preferences show in a technical decision-making context. The results did not support the notion that XAI format alone can significantly influence trust. Instead, individual differences are a more likely determinant. These findings align with Miller (2019), whose theory explains that both cognitive and social factors influence reliance on AI and that XAI should be comprehensible and subjectively satisfying.

Furthermore, this study shows how XAI specificity to its application domain can result in benefits that can be measured qualitatively at participant-level. As the qualitative analysis showed how a SHAP-like explanation (i.e., Feature Importance) can translate PCB design properties into quantitative design scores, which help build a detailed understanding of the AI's internal logic. Additionally, the global explanation of Ideal Design helped build a more beginner-friendly understanding of an AI design agent for PCB making, as the visual and textual elements helped create an intuitive explanation of the AI's goals. Thereby showing the benefit of qualitative evaluation in understanding participants' internal reasoning.

Practical implications can be derived from the qualitative analysis. This study suggests that Ideal Design supports intuitive understanding through concrete and visual explanations, while Feature Importance offers more granular insight into feature-level reasoning, though comes with higher cognitive demand (only supported by qualitative analysis, not by quantitative analysis). This apparent trade-off poses an interesting future direction for XAI research and development, namely by combining these

strengths into a hybrid explanation. Potentially by using the visual overview of Ideal Design, and allowing users to mouse-click through PCB design features to see a specific feature's contribution to the AI's internal design score. Thereby making the explanation more interactive. Another potential direction is to focus on customizable explanations. This study showed strong differences in individual preferences, a future XAI system could adapt to fit a user's preferences and so universally improve user experience. Though the influences of this on performance should be validated.

And methodologically, this study shows the value of combining a wide range of measurements in exploratory studies by uncovering individual preferences that were not captured by questionnaires and behavioral measures. This aligns with recommendations for the HAI and XAI research fields by Kim et al. (2024) and Lai et al. (2023) by evaluating XAI through human-centered frameworks.

Finally, the study demonstrates the value of expanding XAI research into the engineering domain by focusing on a different type of task, explanation method, and user group. Engineering tasks have characteristics unique in the AI research field, and by studying this, new aspects of AI-assisted decision-making can be uncovered.

Limitations

The generalizability of the results is limited by characteristics of the recruited participant sample. Participants were selected via the online platform Prolific (n.d.), with screening based on educational level, relevant work experience, and a custom pre-screening questionnaire to assess domain knowledge. However, observed behavioral data raised concerns about the actual level of electronics engineering expertise in the sample. Specifically, before the feedback intervention, participants were generally unable to identify bad design suggestions. Even though it was confirmed in the pilot study that these bad design suggestions should be obvious to a student engineer. Additionally, the sample had an approximately equal ratio of females and males, which does not reflect the male-dominated demographics of the electronics engineering profession. These findings demonstrate a limit in generalizability to a professional PCB engineering environment. Nonetheless, qualitative insights of this study were valuable in understanding how different XAI methods are experienced in the context of PCB design, thereby contributing to the research aims.

Another limitation comes from the handcrafted AI systems. Instead of experimenting with a functional AI model that gives real design suggestions and calculates its explanations, all were handcrafted by the researcher to simulate AI-generated outputs. This approach was intentional due to the restricted access to professional PCB design software that supports AI-generated designs. However, this approach limited the experiment's design to one wherein participants could only directly interact with the PCB design through a control interface, and each trial was detached from the previous trial by focusing on isolated sections of the PCB. In reality, PCB engineers are more likely to work in a specific order, for example, by routing the important data signals before focusing on the power distribution. In a real-world scenario, interaction with the PCB would be more flexible and logical, and the AI-generated suggestions could have different variability, error rates, capabilities, or interaction methods. Even still, the simulated setup allowed for a controlled environment in which the core research aims could be studied meaningfully.

An order effect is possible because participants always started with the control condition. Different behavior might emerge if the control condition had been queued after the XAI conditions, or if a between-subjects design had been used. However, due to feasibility constraints, the current approach was chosen while remaining aware of any order effects in the discussion. The condition order may have influenced participants' baseline perceptions of the task, potentially influencing the perceived value of the subsequent XAI conditions.

Finally, a lot of variation in this study's models is attributed to individual variation, but could not be explained by the participants' years of relevant experience. This could be a result of missing relevant participant-level predictors. As Vereschak et al. (2021) explained, personal background plays a vital role in shaping user experience, possible predictors could have been AI familiarity or attitude towards automation. Future research could include these predictors to explain some of the variation found in this study.

Recommendations

While the XAI condition had no significant effect on decision rates and self-reported trust in this study, significant individual differences were found, which may be possible to be explained by other

unmeasured factors. Future research should examine participant-level variables like prior AI experience, confidence in self and AI, and perceived vulnerability. Alternatively, task-level elements could be explored, such as users' control over the AI suggestions, decision feedback elements, and quality of AI design suggestions. Including these variables may show a more complete picture of how these XAI methods influence trust in AI and decision rates.

Future studies should take into account the limitations of simulated environments and online recruitment. While this study used a handcrafted AI system to isolate the effects of explanation format, this approach does not capture how users interact with live, adaptive AI systems in a professional engineering environment. Potential new findings can be found when studying with an experiment that is closer to a real-world setting. In other words, following the application-grounded approach as defined by Doshi-Velez & Kim (2017). Potential paths for future research include field experiments in a professional environment, collaborations with ECAD software developers, and experimenting in a longitudinal study to find long-term effects. These approaches would enhance the ecological validity of XAI research directed at specific domains.

Conclusion

This study demonstrated that the inclusion and format of AI explanations significantly influence how users engage with AI assistance in Computer-Aided Design tasks. In an AI-assisted PCB layout task, the local feature-based explanation increased rejection rates of bad design suggestions, indicating supported critical thinking. While the global visual-based explanation "Ideal Design" helped users intuitively understand AI reasoning. Importantly, a wide variety of individual preferences were found that play a major role when aligning users, explanations, and application context. By combining qualitative and quantitative measures, this study gave a deeper view of how users make sense of AI assistance.

These findings highlight the need for personalized hybrid AI explanations in the engineering domain. They show that a single explanation format cannot serve all users and contexts equally. For researchers, the study offers a domain-specific contribution by demonstrating a novel evaluation task and explanation format for Computer-Aided Design tasks. For developers, it suggests that personalized hybrid explanations may offer better results for effective human-AI collaboration.

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Appendix A

Power Analysis – A Priori

The required sample size was calculated using R and the Superpower package. The R-code and output can be seen below:

```
library(Superpower)

mu <- c(5, 5, 5.5)

sd <- c(1.53, 1.53, 1.53)

r <- 0.5

n <- 100

design <- "3w"

labelnames <- c("XAI", "SHAP", "Counterfactual", "OptimalDesign")

design_result <- ANOVA_design(design = "3w",
                             n = n, mu = mu, sd = sd, r = r,
                             labelnames = labelnames)

simulation_result <- ANOVA_power(design_result,
                                 alpha_level = 0.05,
                                 nsims = 1000, verbose = FALSE)
```

Figure A 1

Power Analysis Results

Power and Effect sizes for ANOVA tests		
	power	effect_size
anova_XAI	92.2	0.2498

Power and Effect sizes for pairwise comparisons (t-tests)		
	power	effect_size
p_XAI_SHAP_XAI_Counterfactual	4.3	-0.006175
p_XAI_SHAP_XAI_OptimalDesign	89.1	0.680468
p_XAI_Counterfactual_XAI_OptimalDesign	89.7	0.684957

Appendix B

Screenshots of Experiment Interface

In their invitation through email or by advertising on Prolific, participants were directed to the welcome page (Figure B 1). The figure shows the email login page, for Prolific participants, it would be identical, except they logged in using their Prolific ID.

Figure B 1

Welcome Page With Instructions to Sign Up, Log In, and Find Support

Welcome to the AI-Assisted Electronics Engineering Experiment

In this experiment, you will evaluate the potential role of explainable AI for engineers making electronic circuit designs.
You will work on designing part of a printed circuit board (PCB) using an AI tool.
The experiment takes around 50 minutes and requires your full attention. Scroll down to get started!

1. Sign Up to Participate
Fill out the form below to receive your unique login credentials (if you have not received these yet).
After submitting, your username and password will be shown once only — please save them (take a photo or write them down).

Enter your email

Sign Up

2. Log In to Begin
Only log in if you have 50 minutes available to complete the experiment.
The session requires your full attention and cannot be paused or restarted once started.

Enter your email

Password

Start Experiment

Support & Questions
If you encounter issues with logging in, lost your credentials, or have questions about the experiment, feel free to contact us:

Email: t.p.bruinsma@student.tue.nl
WhatsApp: [+31 6 41533398](https://wa.me/31641533398)

Conducted by Theo Bruinsma, Eindhoven University of Technology

Logging in directed participants to the pre-screening questionnaire (Figure B 2). They had three minutes to complete multiple-choice questions specific to the PCB engineering domain.

Figure B 2

Pre-Screening Page with a Repetition of the Target Group, Timer, and Questions

Pre-Screening Questionnaire

This experiment is only intended for people with experience related to circuit board design.
Usage of PC or laptop is required, mobiles won't work correctly.

1. What is the primary function of a decoupling capacitor placed close to an ICs power pin?

☐ To store charge for later use during low-voltage events
☐ To equalize current through the power rail
☐ To increase overall PCB capacitance
☐ To minimize voltage ripple and suppress high-frequency noise
2. What is the risk of routing a high-speed digital signal in parallel to another trace over a long distance?

☐ The trace will radiate excessive heat
☐ The signal will experience thermal drift
☐ The signal will experience crosstalk
☐ Signal propagation will slow down significantly
3. Why is a continuous ground plane beneficial in a multilayer PCB?

☐ It increases board rigidity and prevents warping
☐ It helps reduce EMI and maintain consistent signal return paths
☐ It eliminates the need for decoupling capacitors by stabilizing voltage directly beneath the IC
☐ It acts as a backup power supply during transients
4. Why is it critical to match the land pattern (footprint) of a component in the PCB design phase?

☐ To allow automated test probes to access the component during functional testing
☐ To ensure proper soldering and alignment
☐ To reduce the number of vias required
☐ To allow for interchanging components in assembly
5. Which design strategy reduces inductance and improves high-speed signal integrity?

☐ Placing signal traces directly over continuous ground planes
☐ Minimizing the use of ground planes to avoid parasitic capacitance
☐ Using long, meandering traces to match high-speed frequencies
☐ Routing signals through multiple vias to reduce resistance

03:51

Participants who qualified for the study were shown the informed consent form (Figure B 3). Participants could explicitly state if they agreed to having their data shared in the Open Science Framework public repository and whether they consented to participating in the study.

Figure B 3

Informed Consent Page

Information Sheet

Research Project: A Comparative Study of Explainable AI Methods for PCB Design Tasks

Introduction

You have been invited to take part in research project "A Comparative Study of Explainable AI Methods for PCB Design Tasks". We are asking you because you are an Electrical Engineering student with experience or interest in printed circuit board (PCB) design. Your input will help us understand how engineers make design choices and how AI tools that provide explanations can support that process.

Participation in this research project is voluntary: the decision to take part is up to you. Before you decide to participate we would like to ask you to read the following information, so that you know what the research project is about, what we expect from you and how we deal with processing your personal data. Based on this information you can indicate via the consent declaration whether you consent to take part in this research project and the processing of your personal data.

Purpose of the Research

This research project will be managed by Theo Bruinma under supervision of Dr. Chao Zhang and Prof. Dr. Chris Snijder of the Human-Technology Interaction group. The purpose of this research project is understand which explanations for AI-generated PCB design choices you prefer. The data collected in this study will be used to improve future PCB Design Software and their AI integration.

Controller in the Sense of the GDPR

TU/e is responsible for processing your personal data within the scope of the research.

Contact details:

Technische Universiteit Eindhoven
De Groene Loper 3
5612 AE, Eindhoven

What Will Taking Part in the Research Project Involve?

You will be taking part in a research project in which we will gather information by:

- Measuring your behavior during PCB design tasks.
- Asking you to fill in questionnaires about your opinion of the AI.
- Asking you to fill in a questionnaire about your demographics.

Potential Risks and Inconveniences

Your participation in this research project does not involve any physical, legal or economic risks. You do not have to answer questions which you do not wish to answer. Your participation is voluntary. This means that you may end your participation at any moment you choose by letting the researcher know this. You do not have to explain why you decided to end your participation in the research project. None of this will have any negative consequences for you whatsoever.

What Personal Data Will We Gather and Process?

Contact Data: Age, gender **Engineering Profile:** Educational level, years of experience in electronics

Confidentiality, Use, and Storage of Data

We will do everything we can to protect your privacy as best as possible. The research results that will be published will not in any way contain confidential information or personal data from or about you through which anyone can recognize you, unless in our consent form you have explicitly given your consent for mentioning your name, for example in a quote. The personal data that were gathered via offline surveys and behavior measures. This data and other documents within the framework of this research project, will be stored on an encrypted server of the Human-Technology Interaction group. The raw and processed research data will be retained for a period of 10 years. The research data will, if necessary (e.g. for a check on scientific integrity) and only in anonymous form be made available to persons outside the research group. If you give your consent, your data will be made available for future research, for example by sharing those data with fellow researchers or making anonymized/pseudonymized data available via an open access data archive or repository.

Withdrawing Your Consent and Contact Details

Participation in this research project is entirely voluntary. You may end your participation in the research project at any moment, or withdraw your consent to using your data for the research, without specifying any reason. Ending your participation will have no disadvantageous consequences for you.

Do you wish to end the research, or do you have any questions and/or complaints? Then please contact the researcher Theo Bruinma via t.bruinma@student.tue.nl. Alternatively you may contact the researcher's supervisor c.zhang@tue.nl. Or alternatively:

- Questions regarding privacy topics, mail the TU/e privacy officer: privacy@tue.nl
- Questions regarding ethical topics, mail the TU/e Ethical Review Board secretary: ethics@tue.nl

Legal Ground for Processing Your Data

The legal basis upon which we process your data is consent. This research project was assessed and approved by the ethical review committee of the HTI group of the Eindhoven University of Technology.

Consent Form for Participation

By signing this consent form, I acknowledge the following:

1. I am sufficiently informed about the research project through a separate information sheet. I have read the information sheet and have had the opportunity to ask questions. These questions have been answered satisfactorily.
2. I take part in this research project voluntarily. There is no explicit or implicit pressure for me to take part in this research project. It is clear to me that I can end participation in this research project at any moment, without giving any reason. I do not have to answer a question if I do not wish to do so.

Furthermore, I consent to the following parts of the research project:

Certificate of Consent

I consent to making research data gathered from me available via a repository, namely the "Open Science Framework", for the purpose of retaining the data and using it for future research as described above.

☐ YES ☐ NO

Certificate of Consent

I consent to participate in this study.

☐ YES ☐ NO

[Continue](#)

After signing the consent form, they were shown a video tutorial. They could only continue the experiment after they had waited the full duration of the video.

Video tutorial – part 1: <https://youtu.be/0fkHmXtWBAw>

Participants were in the first condition and were shown the PCB engineering task without the XAI method (Figure B 4). The interface included a control interface to explore, accept, and reject the AI suggestions. Additionally, the info button allowed participants to view the technical background of PCB design, which included a block diagram, schematic, 3D renders, color legend, and an unrouted PCB layout (Figure B 6). After the control condition (i.e., 10 PCB engineering tasks, 3 error detection tasks, and the trust questionnaire), participants were directed to a second video tutorial to explain the XAI conditions.

Video tutorial – part 2: <https://youtu.be/d29zPwkfNt8>

Figure B 4

PCB Layout Task, Control Condition Without XAI

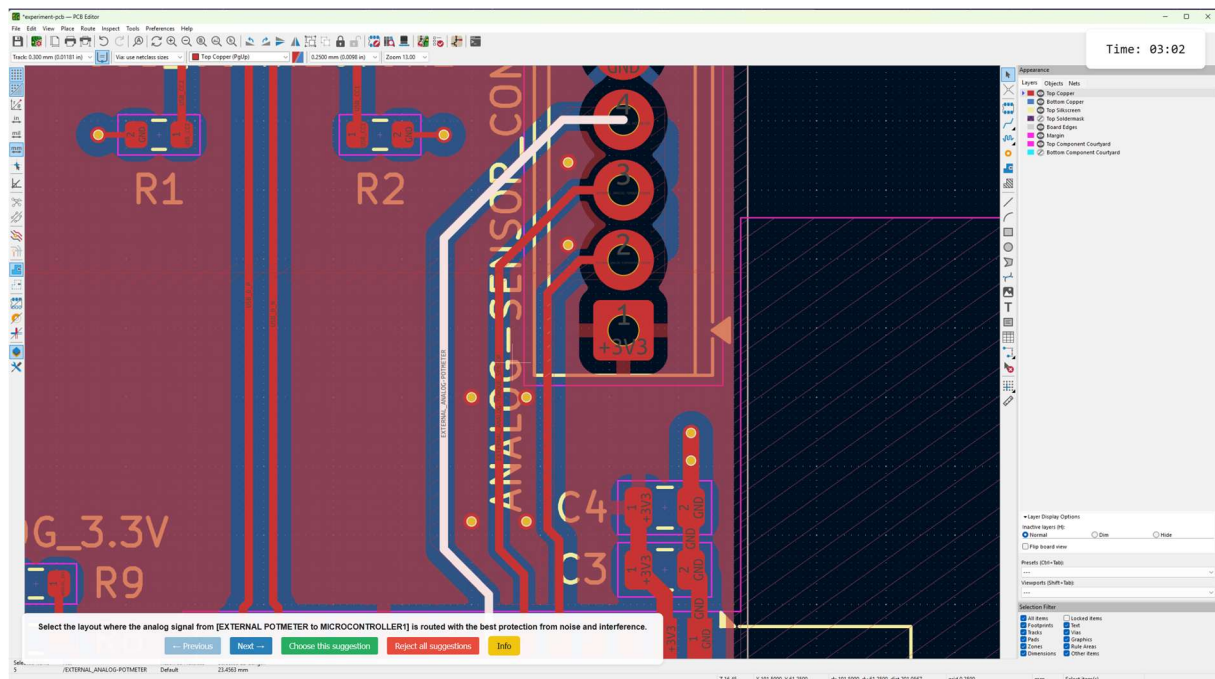
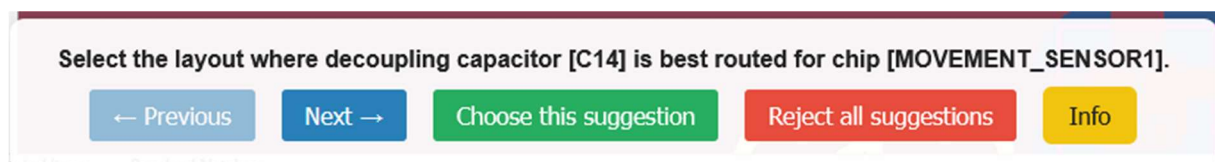


Figure B 5

Control Interface Shown at the Bottom of the Screen During PCB Engineering Tasks



After finishing the second video tutorial, participants were shown PCB engineering tasks, this time, however, with an Ideal Design explanation (Figure B 7) or the Feature Importance explanation (Figure B 8).

Figure B 7

PCB Layout Task, Ideal Design Condition

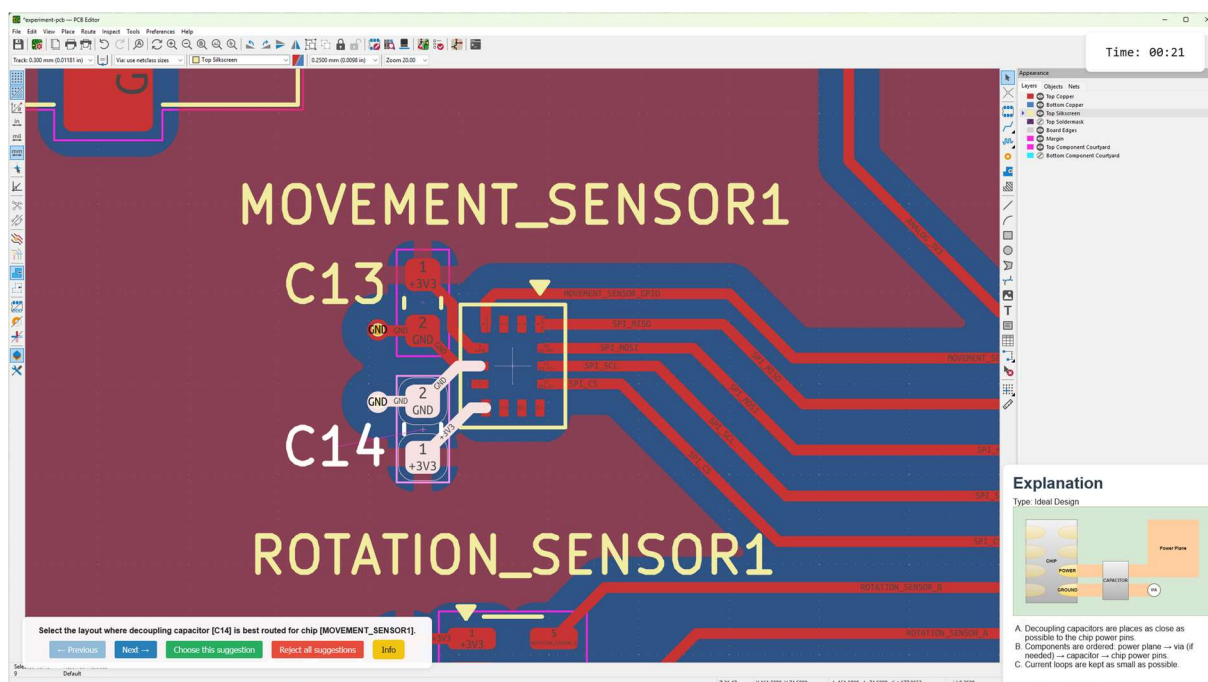
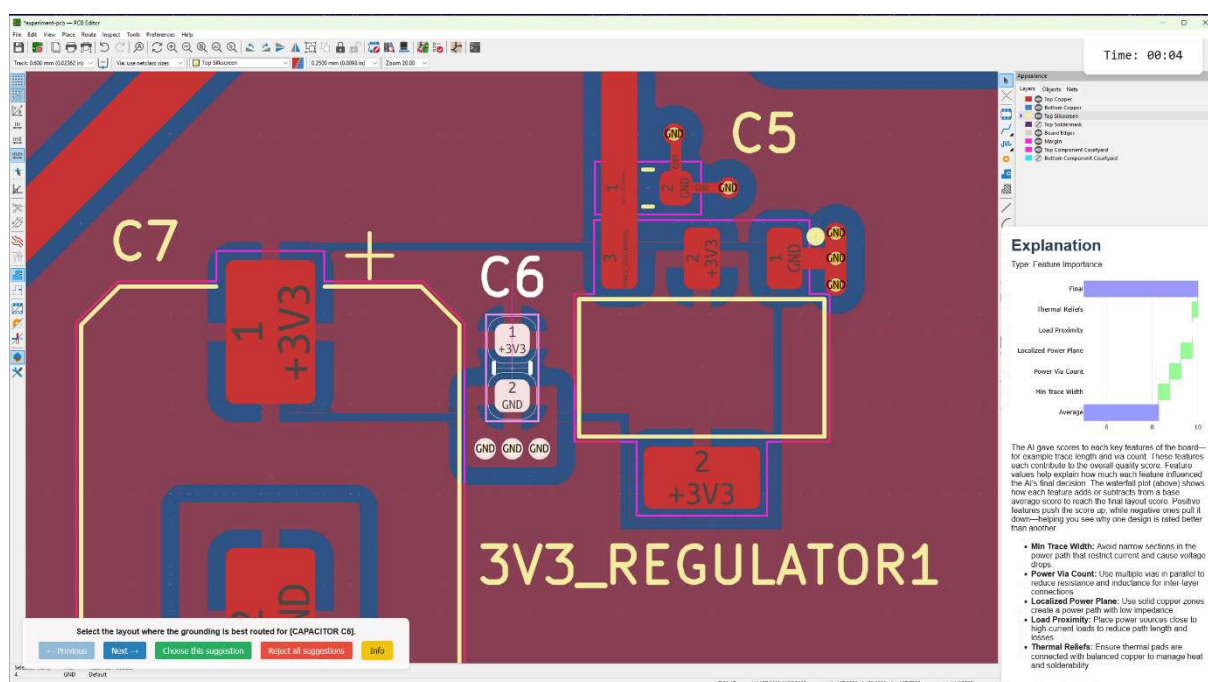


Figure B 8

PCB Layout Task, Feature Importance Condition



Questionnaires were shown after each condition (Figure B 9).

Figure B 9

Questionnaire Pages for Mental Effort, Satisfaction and Trust (all were Shown on Separate Webpages)

Questionnaire

Mental Effort

How much mental and perceptual activity was required? Was the task easy or demanding, simple or complex?

Very Low Very High 10

Finish

Questionnaire

Satisfaction

From the explanation, I know how the AI designer works.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

This explanation of how the AI designer works is satisfying.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

This explanation of how the AI designer works has sufficient detail.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

This explanation of how the AI designer works seems complete.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

This explanation of how the AI designer works tells me how to use it.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

This explanation of how the AI designer works is useful to my goals.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

This explanation of the AI designer shows me how accurate the AI designer is.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

Next

Questionnaire

Trust

I am confident in the AI designer. I feel that it works well.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

The outputs of the AI designer are very predictable.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

The AI designer is very reliable. I can count on it to be correct all the time.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

I feel safe that when I rely on the AI designer I will get the right answers.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

The AI designer is efficient in that it works very quickly.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

I am wary (cautious) of the AI designer.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

The AI designer can perform the task better than a novice human user.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

I like using the AI designer for decision making.

☐ Strongly Agree
☐ Agree
☐ Neither Agree nor Disagree
☐ Disagree
☐ Strongly Disagree

Next

After all conditions and questionnaires were completed, a final questionnaire was shown to gather preferences, qualitative feedback, and demographics Figure B 10.

Figure B 10

Questionnaire Pages for Preferences, Qualitative Feedback, and Demographics (all were Shown on a Singular Webpage)

Conclusion

Feature Importance

Explanation
Type: Feature Quality

The Feature Importance explanation highlights which features of the PCB design were most influential in the AI's decision-making process. This helps you understand what aspects the AI considered most important for its suggestions.

Ideal Design

Explanation
Type: Ideal Design

The Ideal Design explanation shows an example of a highly-rated PCB design, illustrating what the AI considers to be an optimal solution. This provides a concrete reference for what the AI is aiming for.

Feature Importance

What did you like or find helpful about the AI explanation Feature Importance?

What did you dislike or find unhelpful about the AI explanation Feature Importance?

Demographics

What's your age?

What's your gender?
☐ Male ☐ Female ☐ Other

What's your highest (completed) educational degree relevant to electronics?
☐ Bachelor ☐ Master ☐ Not graduated yet ☐ Other

How many years of experience do you have in electronics (including educational years)?

Ideal Design

What did you like or find helpful about the AI explanation Ideal Design?

What did you dislike or find unhelpful about the AI explanation Ideal Design?

Which explanation method (see images above) do you prefer?

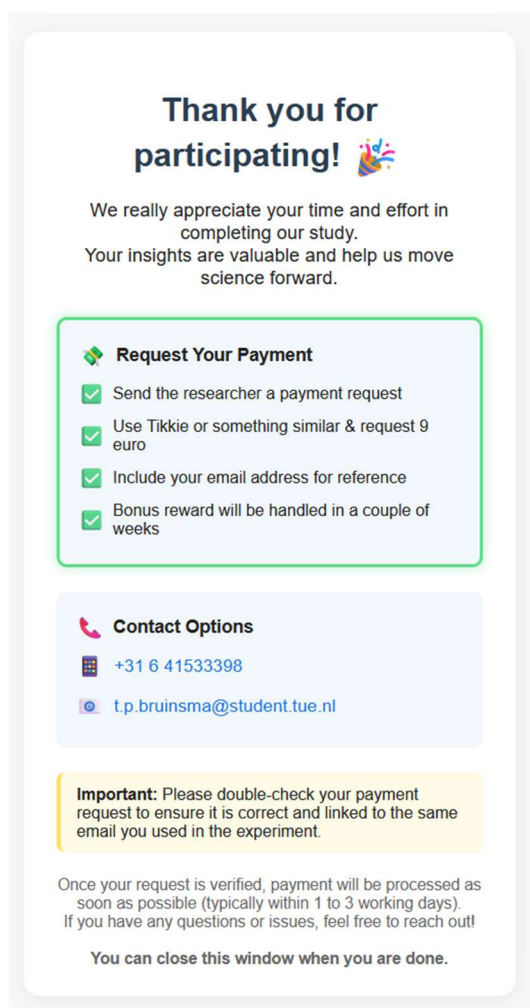
☐ Feature Importance
 ☐ Ideal Design

Submit

After completing the final questionnaire, TU/e participants were directed to the closing page, which included payment instructions and contact information. Prolific participant payments were handled through the Prolific website.

Figure B 11

Payment Page with a Thank You Message, Payment Instructions, and Contact Information



Thank you for participating! 🎉

We really appreciate your time and effort in completing our study.
Your insights are valuable and help us move science forward.

Request Your Payment

- ✓ Send the researcher a payment request
- ✓ Use Tikkie or something similar & request 9 euro
- ✓ Include your email address for reference
- ✓ Bonus reward will be handled in a couple of weeks

Contact Options

+31 6 41533398

t.p.bruinsma@student.tue.nl

Important: Please double-check your payment request to ensure it is correct and linked to the same email you used in the experiment.

Once your request is verified, payment will be processed as soon as possible (typically within 1 to 3 working days).
If you have any questions or issues, feel free to reach out!

You can close this window when you are done.

Appendix C

Questionnaire Content

Pre-Screening (multiple choice)

The following multiple-choice questions were used to assess participants' baseline knowledge relevant to PCB design. Correct answers are indicated with [Correct answer].

1. What is the primary function of a decoupling capacitor placed close to an IC's power pin?
 - To store charge for later use during low-voltage events
 - To equalize current through the power rail
 - To increase overall PCB capacitance
 - [Correct answer] To minimize voltage ripple and suppress high-frequency noise
2. What is the risk of routing a high-speed digital signal in parallel to another trace over a long distance?
 - The trace will radiate excessive heat
 - The signal will experience thermal drift
 - [Correct answer] The signal will experience crosstalk
 - Signal propagation will slow down significantly
3. Why is a continuous ground plane beneficial in a multilayer PCB?
 - It increases board rigidity and prevents warping
 - [Correct answer] It helps reduce EMI and maintain consistent signal return paths
 - It eliminates the need for decoupling capacitors by stabilizing voltage directly beneath the IC
 - It acts as a backup power supply during transients
4. Why is it critical to match the land pattern (footprint) of a component in the PCB design phase?
 - To allow automated test probes to access the component during functional testing
 - [Correct answer] To ensure proper soldering and alignment
 - To reduce the number of vias required
 - To allow for interchanging components in assembly

5. Which design strategy reduces inductance and improves high-speed signal integrity?
- [Correct answer] Placing signal traces directly over continuous ground planes
 - Minimizing the use of ground planes to avoid parasitic capacitance
 - Using long, meandering traces to match high-speed frequencies
 - Routing signals through multiple vias to reduce resistance

Satisfaction (5-Point Likert Scale) (Hoffman et al., 2023)

1. From the explanation, I know how the AI designer works.
2. This explanation of how the AI designer works is satisfying.
3. This explanation of how the AI designer works has sufficient detail.
4. This explanation of how the AI designer works seems complete.
5. This explanation of how the AI designer works tells me how to use it.
6. This explanation of how the AI designer works is useful to my goals.
7. This explanation of the AI designer shows me how accurate the AI designer is.

Trust (5-Point Likert Scale) (Hoffman et al., 2023)

1. I am confident in the AI designer. I feel that it works well.
2. The outputs of the AI designer are very predictable.
3. The AI designer is very reliable. I can count on it to be correct all the time.
4. I feel safe that when I rely on the AI designer I will get the right answers.
5. The AI designer is efficient in that it works very quickly.
6. I am wary (cautious) of the AI designer.
7. The AI designer can perform the task better than a novice human user.
8. I like using the AI designer for decision making.

Mental Effort (Hart & Staveland, 1988)

How much mental and perceptual activity was required? Was the task easy or demanding, simple or complex?

Preference

Which explanation method do you prefer?

☐ Feature Importance ☐ Ideal Design

Qualitative Feedback

What did you like or find helpful about the AI explanation Ideal Design?

What did you dislike or find unhelpful about the AI explanation Ideal Design?

What did you like or find helpful about the AI explanation Feature Importance?

What did you dislike or find unhelpful about the AI explanation Feature Importance?

Demographics

What's your age?

What's your gender?

What's your highest (completed) educational degree relevant to electronics?

How many years of experience do you have in electronics (including educational years)?

Appendix D

Correlation Matrix

Table D 1

Correlation Matrix for Subjective and Objective Measures

	Control		Feature Importance			Ideal Design			
	Trust	Trust	Satisfaction	Mental effort	Preference	Trust	Satisfaction	Mental Effort	Preference
Decision time	.263	.087	.167	.090	-.119	.216	.246	.001	.051
Final suggestion	.120	-.113	.061	.168	-.279	-.048	-.059	.151	-.069
Level of questioning	.088	-.109	.027	.083	-.103	-.108	-.099	.148	-.024
First condition is Ideal Design	.080	-.066	-.090	-.149	.243	.004	-.038	-.028	-.243
Intervention included	-.068	-.076	-.004	.038	-.248	-.144	-.057	.073	.248
Acceptance of good designs	.277	.085	.182	.162	-.273	.286	.255	-.066	.222
Rejection of bad designs	-.378*	-.088	-.140	-.163	.365*	.033	.013	.069	-.272

Note. * $p < .05$

Appendix E

Decision Time Throughout the Experiment

Figure E 1

Margins Plots of Decision Time for Both Condition Orders, Showing Potential Learning Effects

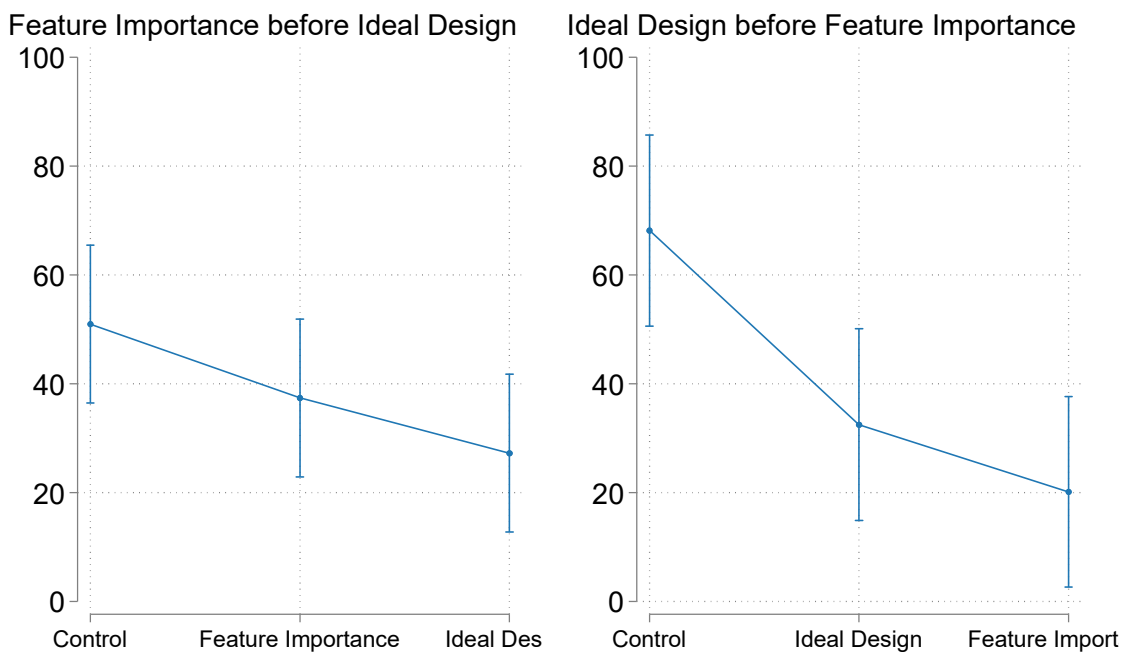
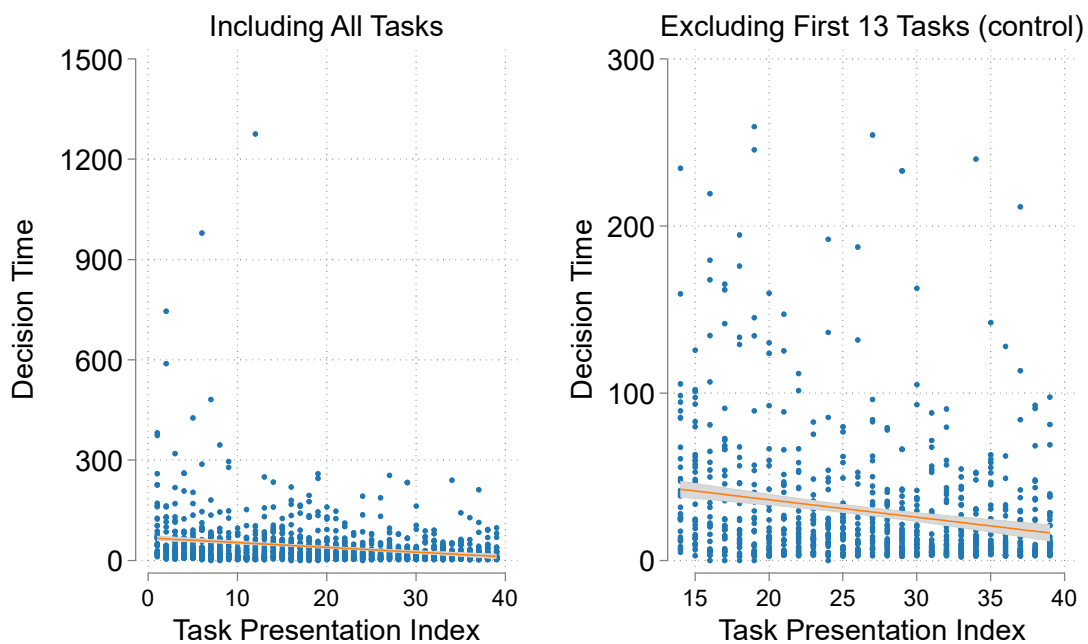


Figure E 2

Box Plots of Decision Time on Trial-Level With Linear Fit and 95% Confidence Interval



Appendix F

Summarized Answers to Open-Ended Questions

Table F 1

Feature Importance, Helpful Aspects Summarized

Helpful aspects	Summary of Participant Feedback	Frequency
Feature Importance and prioritization	Liked that the explanation showed which PCB design aspects influenced AI decisions or scores, helping them understand what to focus on or improve.	9
Comprehensibility and clarity	Appreciated clear, understandable explanations or displays that improved trust or mental processing.	7
Visual presentation and structure	Found graphs, diagrams, or clear layout visually helpful in interpreting AI output.	4
Decision support for PCB improvement	Found the explanation useful for making design decisions, improving outcomes, or guiding the design process.	5
Process explanation	Liked how the explanation clarified the AI's internal logic, aims, or steps — including real-time reasoning.	4
Efficiency and ease of use	Felt that the explanation made their work easier or quicker.	2

Table F 2*Feature Importance, Unhelpful Aspects Summarized*

Unhelpful aspects	Description	Frequency
Too abstract or difficult to understand	Participants found the explanation hard to understand or overly abstract. They noted a lack of concrete examples or actionable guidance.	9
Missing or unclear visual guidance	Visual aids were confusing, missing, or didn't show layouts or concrete representations.	5
Lacks specificity or explanatory detail	The explanation lacked precision, context, or reasoning behind design decisions.	4
Overly technical or AI focused	Participants felt the explanation focused too much on abstract graphs or the AI's internal logic, rather than user-friendly insights.	2
Minor clarity or presentation issues	Small issues with wording, color coding, or unclear phrasing were mentioned.	2
No negative feedback given	Participants explicitly stated that they found no issues or were fully satisfied with the explanation.	10

Table F 3*Ideal Design, Helpful Aspects Summarized*

Helpful aspects	Description	Frequency
Clarity and understandability	Explanation was well-structured, detailed, and easy to understand — even when complex.	8
Concrete examples of good design	Provided a clear reference or example of what a successful layout looks like, aiding interpretation.	8
Visual representation and layout clarity	Diagrams, graphics, and visualizations helped participants see how the design works or should look.	7
Helps improve design quality and efficiency	Made it easier to design, decide, or iterate faster and more accurately.	5
Insight into key features	Clarified which features mattered most in the AI's scoring or decision.	2

Table F 4*Ideal Design, Unhelpful Aspects Summarized*

Unhelpful aspects	Description	Frequency
Too vague or generic in feedback	The explanation lacked specificity or clarity in what was wrong or how features impacted results.	4
Lacks justification for AI decisions	Did not explain why a certain layout was preferred or what happens if rules are not followed.	4
Lacks detail or clarity	Not enough detail or precision in the explanation; visuals or language were unclear or confusing.	5
Includes unnecessary information	Some content was irrelevant or too much, reducing usefulness.	2
Missing design score for comparison	The explanation lacked a global efficiency metric, making it harder to compare board suggestions.	1
No negative feedback given	Participants found the explanation entirely helpful or gave no critical comments.	10